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AND TECHNOLOGY

# Dynamic Structural Analysis in Marc Mentat

Materiały i Struktury Inteligentne (2018) – przygotował: Jakub Bryła
opieka merytoryczna: dr inż. Adam Martowicz

# Agenda:

-> Problem description

-> 1st problem:

=> Modal analysis

=> Transient analysis: change of load states during an analysis

-> 2nd problem:

=> Harmonic analysis

=> Transient analysis: applied load dependent of time

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# Problem description

The goal is to find maximum deflection of a beam.

Geometry data: The beam has rectangular cross section with side dimensions 4 and 8mm. The length of the item is 50mm. The beam is made of the steel.

Boundary condition: The beam is fixed on one side, while on the other side a 650N force is applied at center of the cross section, along shorter side.

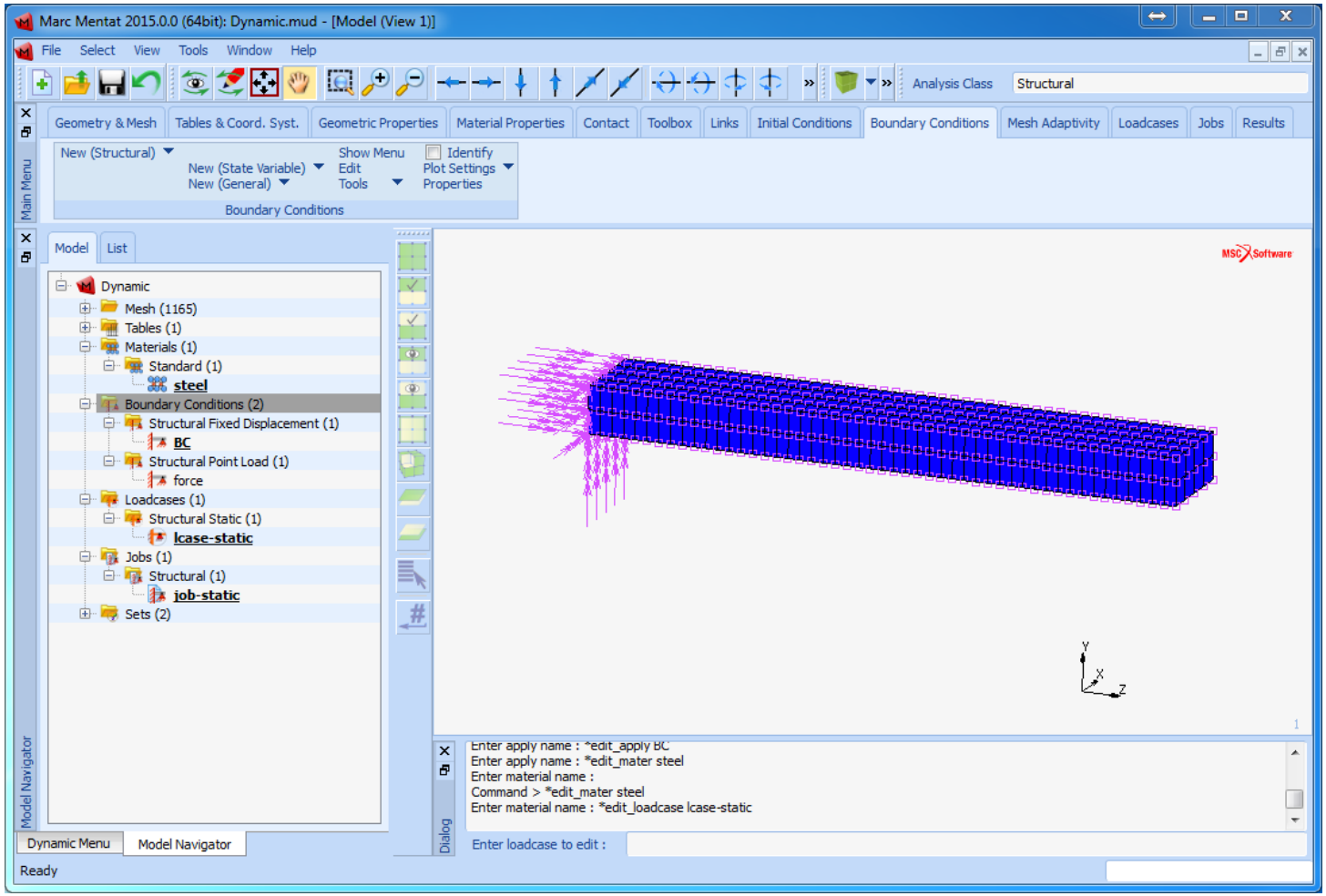
1st problem: After the beam reaches the maximum displacement, the load is released. Analyse deflection of the beam at time.

2nd problem: Apply load as harmonic function with frequency which guarantee maximum deflection.

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# 1st problem: Modal analysis

Prepare model as is described in *Static Structural Analysis in Marc Mentat* tutorial.



# 1st problem: Modal analysis

A material density must be defined for dynamic analyzes.

So far it have been defined forces at Newton and dimensions in mm. Therefore mass must be difened at tones:

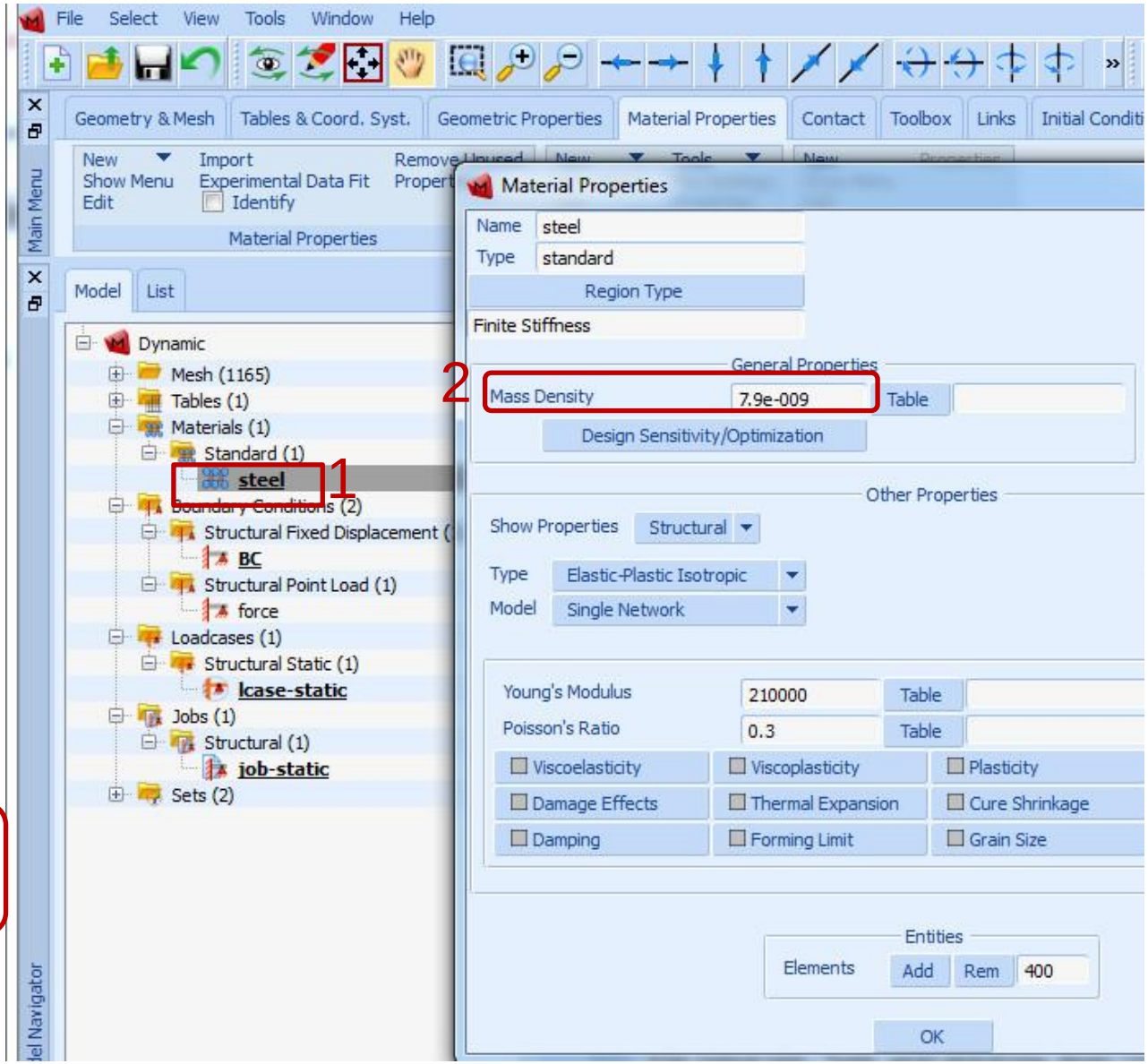
$$N = \frac{kg \cdot m}{s^2} = \frac{kg \cdot (10^3 mm)}{s^2} = \frac{T \cdot mm}{s^2}$$

Therefore the value of steel density will be:

$$7900 \frac{kg}{m^3} = 7.9 \cdot 10^3 \frac{10^{-3} T}{10^9 mm^3} = 7.9 \cdot 10^{-9} \frac{T}{mm^3}$$

Step 1:
Double click on defined material – steel.

Step 2:
Input calculated value of the steel density.






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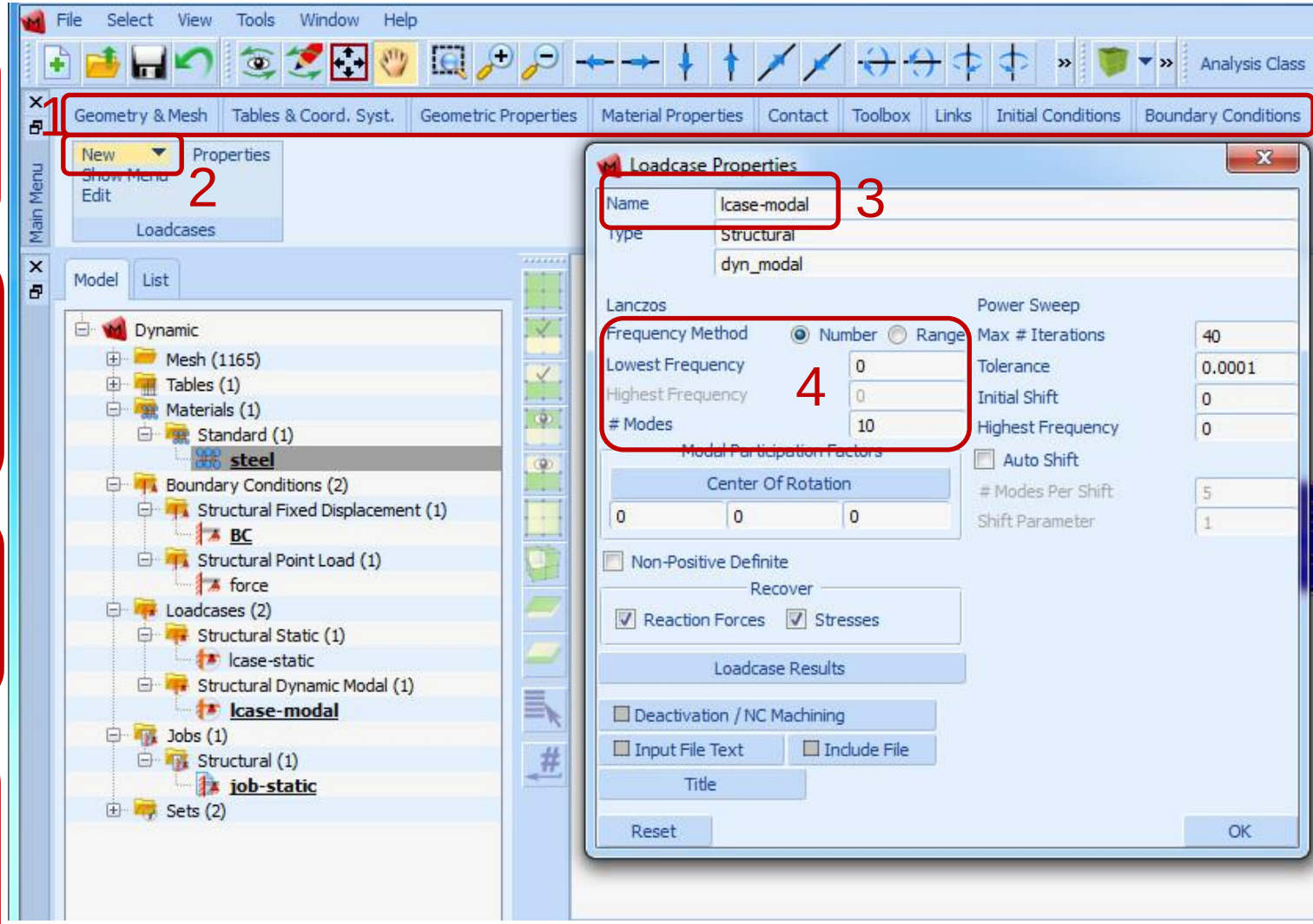
# 1st problem: Modal analysis

Step 1:
Go to *Loadcases* Tab from Main Menu.

Step 2:
Click on *New* list and choose *Dynamic Modal*.

Step 3:
Change *name* to „lcase-modal”.

Step 4:
Input demanded number of modes or range of frequency at which the modes will be searched.






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# 1st problem: Modal analysis

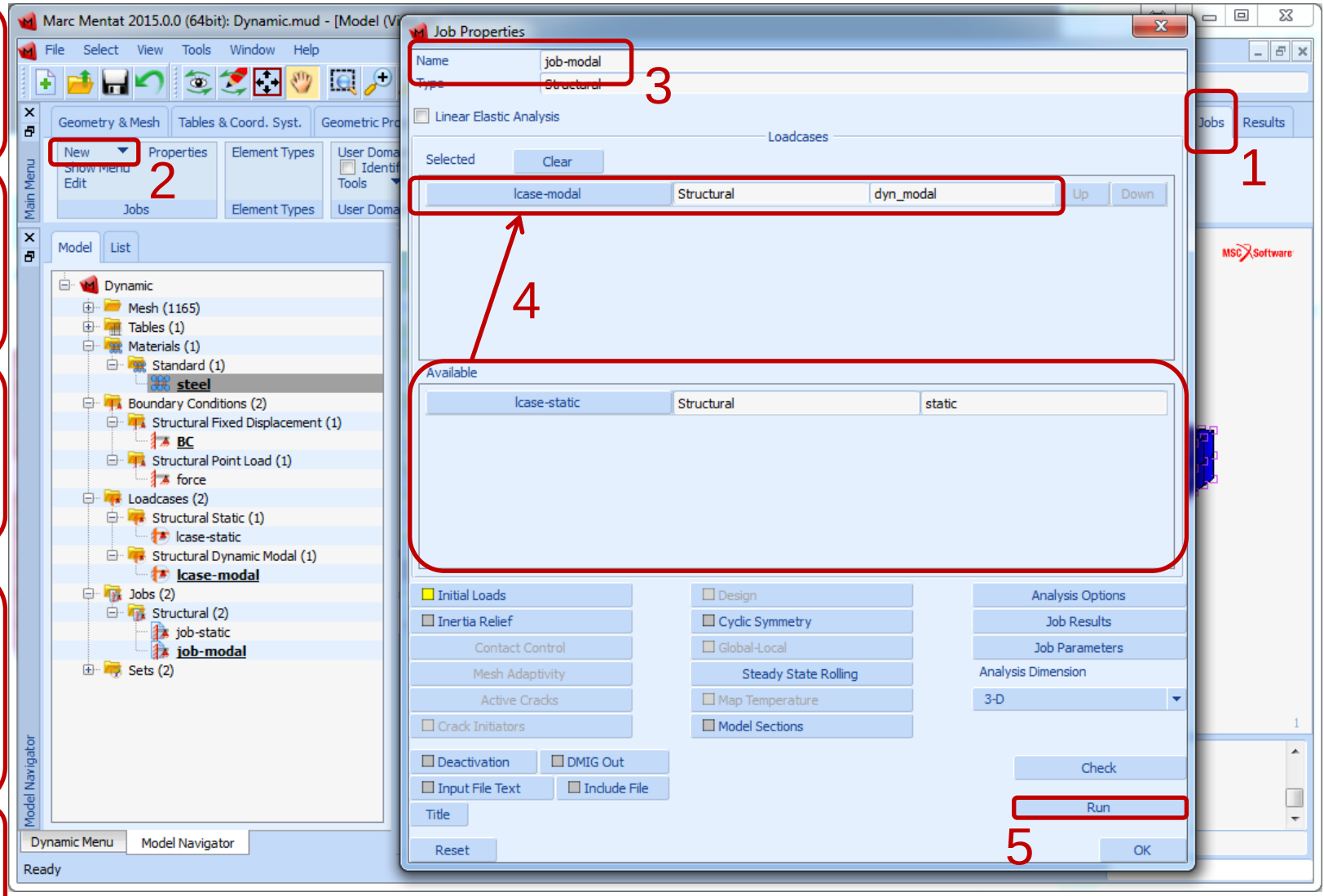
Step 1:
Go to *Job* Tab from *Main Menu*.

Step 2:
Click on *New* and choose *Structural*.

Step 3:
Change *Name* to „job-modal”.

Step 4:
Select *lcase-modal* loadcase from *Available* area.

Step 5:
Click *Run* Button.





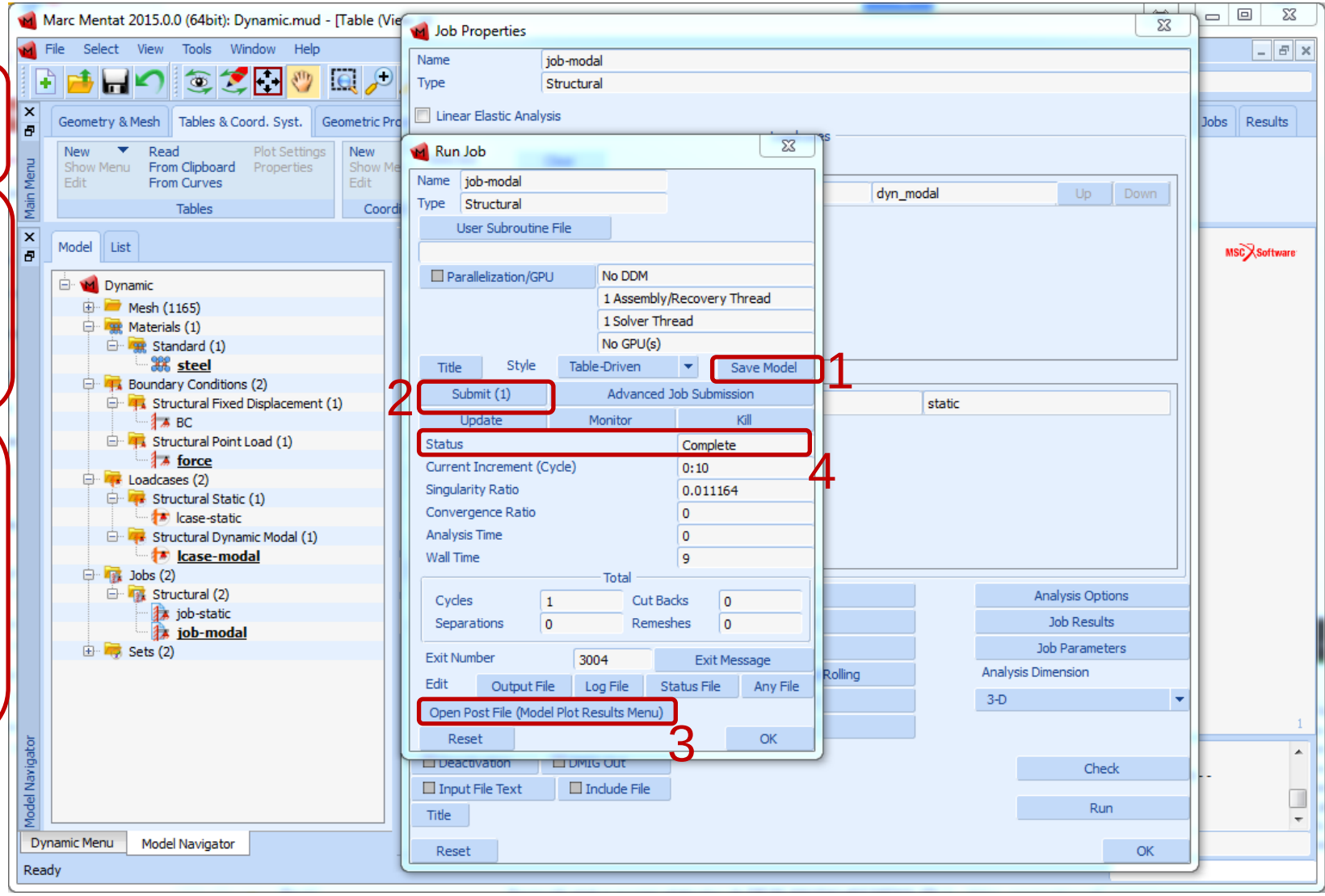

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# 1st problem: Modal analysis

Step 1:
Save model.

Step 2:
Run analysis by clicking *Submit(1)* Button.

Step 3:
Click *Open Post File…* Button after *Status* Bar change to *Complete* 4 .






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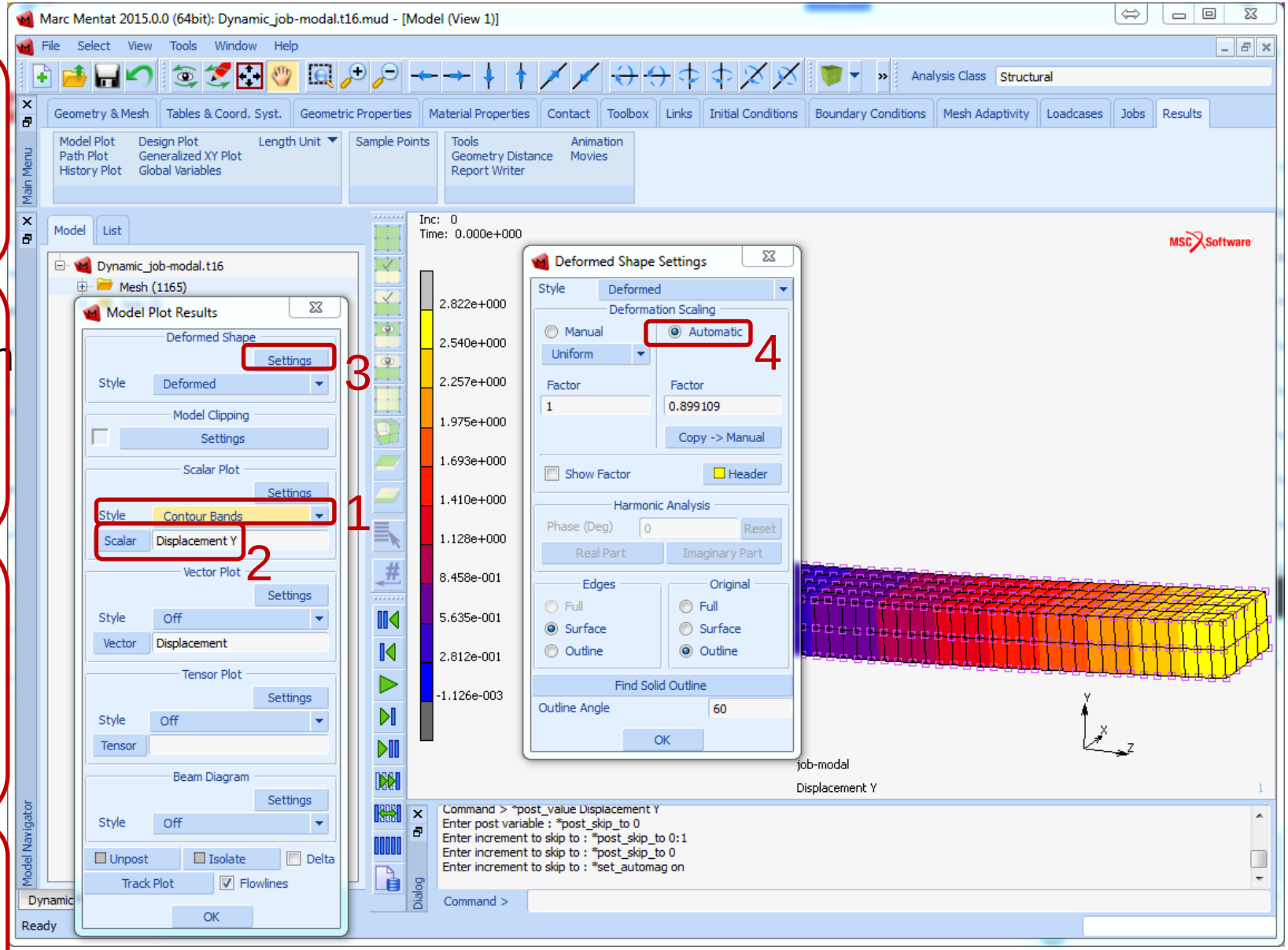
# 1st problem: Modal analysis

Step 1:
Choose *Contour Bands* option from *Style* list.

Step 2:
Click *Scalar* Button and choose *Displacement Y* option.

Step 3:
Click *Settings* Button from *Deformed Shape* Area.

Step 4:
Change *Deformation Scaling* to *Automatic*.





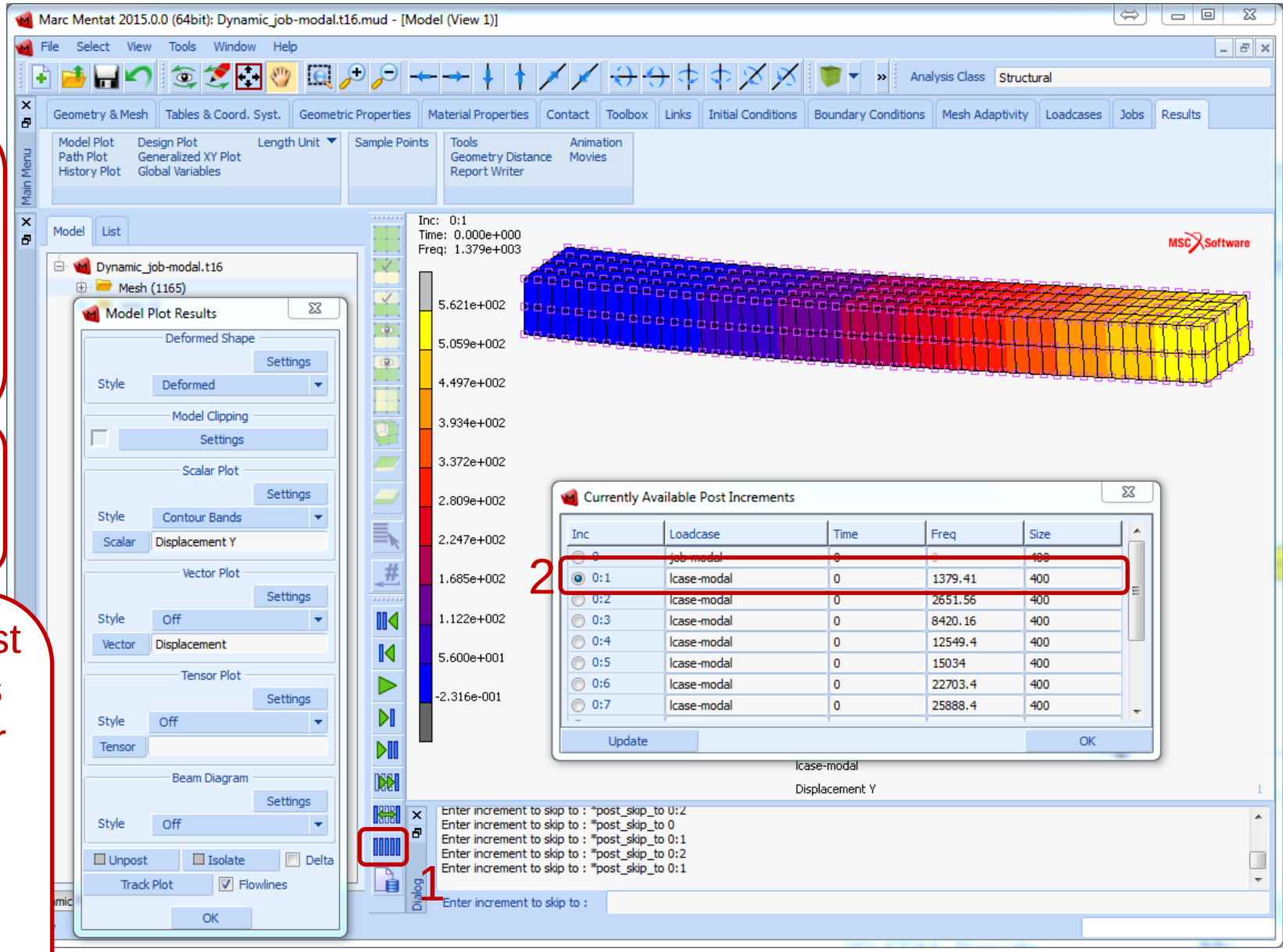

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# 1st problem: Modal analysis

Step 1:
Click *Scan results file* Button to open *Currently Available Post Increments* Window.

Step 2:
Choose the first result.

Observe that the first modal frequency is the wanted one, for which the beam bending along Y axis. Write down the frequency: **1379.41Hz**.

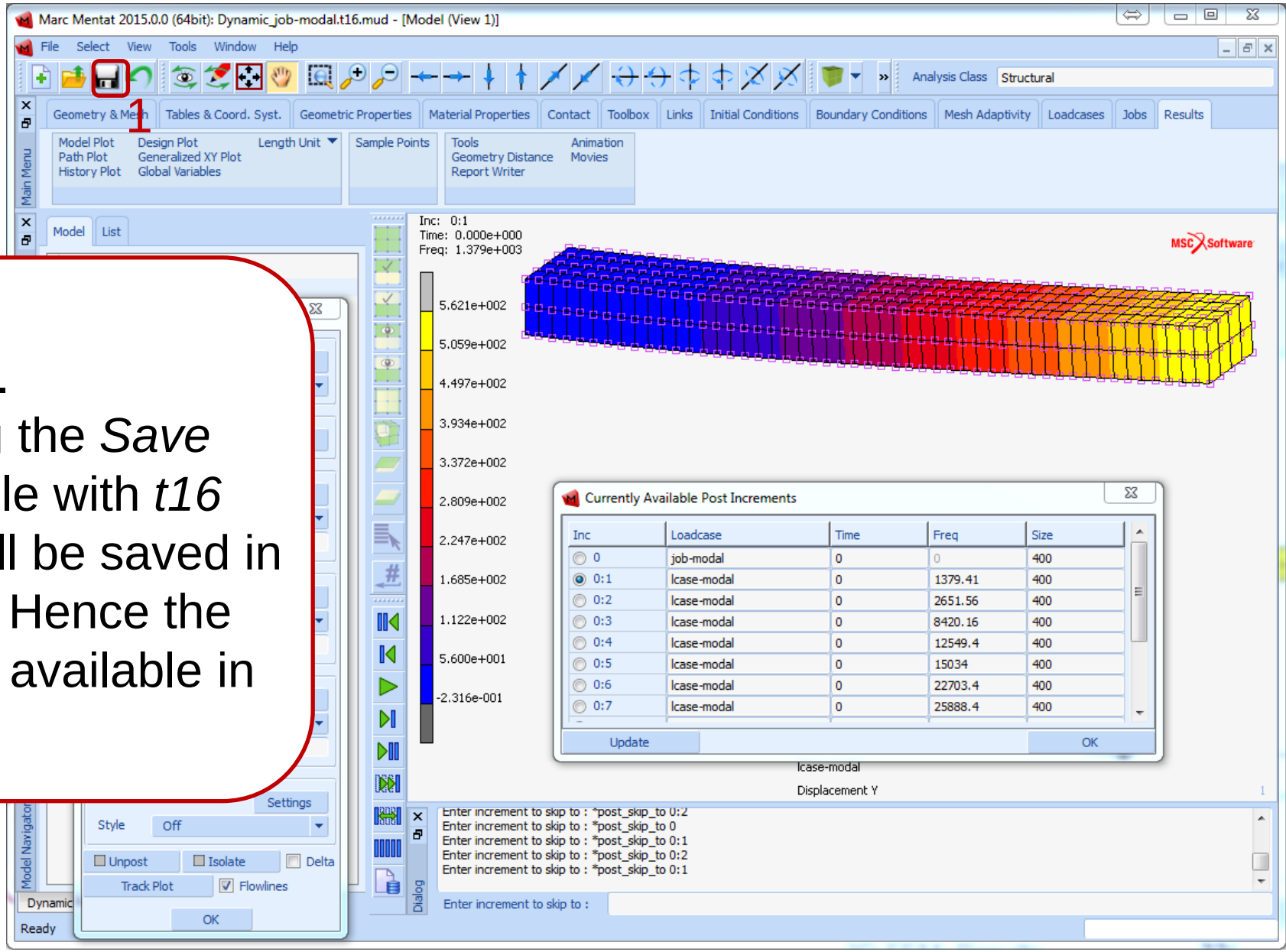





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# 1st problem: Modal analysis

Step 1:
Save results.
After clicking the *Save* Button, the file with *t16* extension will be saved in PC memory. Hence the result will be available in future.






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# 1st problem: Transient analysis (without Damping)

Step 1:
Open model with the Beam.

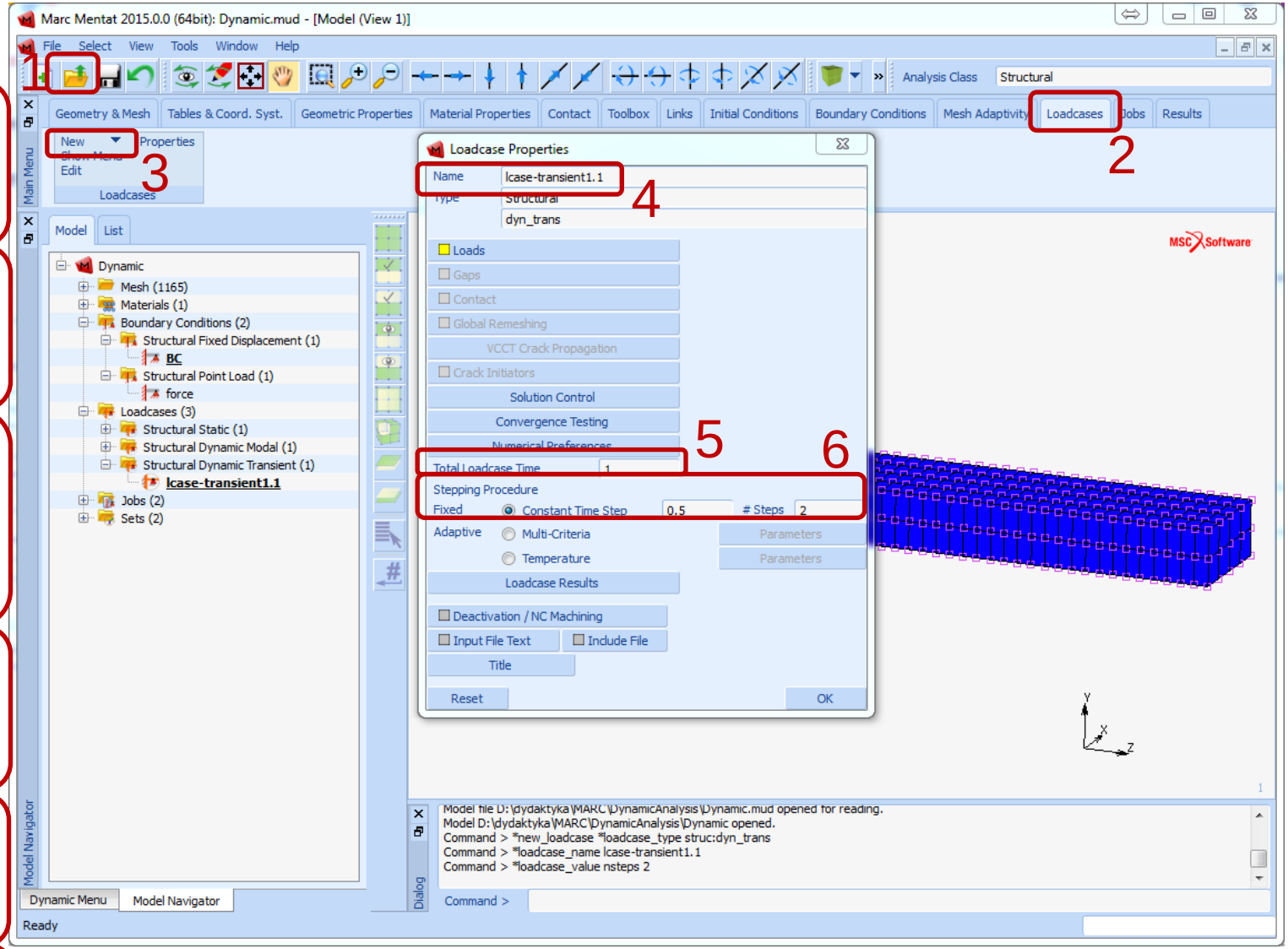
Step 2:
Go to *Loadcases* Tab from *Main Menu*.

Step 3:
Click on *New* list and choose *Dynamic Transient*.

Step 4:
Change Name to „lcase-transient1.1”

Step 5:
Set *Total Loadcase Time* to 1.

Step 6:
Set *Steps* number to 2.






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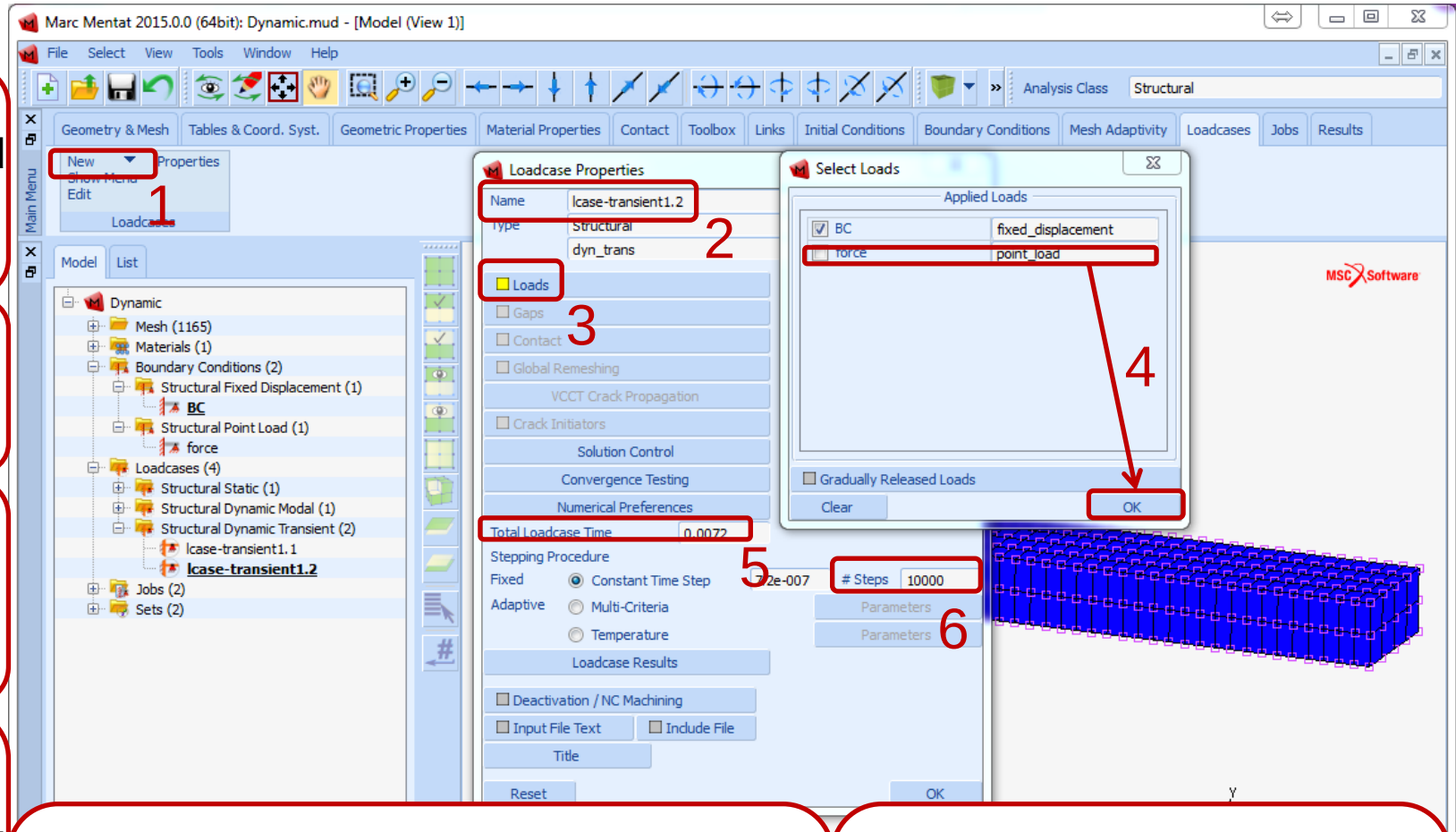
# 1st problem: Transient analysis (without Damping)

Step 1:
Click on *New* list and choose *Dynamic Transient*.

Step 2:
Change Name to „lcase-transient1.2”.

Step 3:
Click *Loads* Button to open *Select Loads* Window.

Step 4:
Unselect *force* from *Applied Loads* Area and click *OK* Button to close *Select Loads* Window.




Step 5:
To observe 10 deflection cycles the Total Loadcase Time must be equal:

$$10 \cdot T = \frac{10}{f} = \frac{10}{1379.41Hz} = 0.0072s$$

Step 6:
Set 2000 steps, what gives 200 samples per one deflection cycle.

# 1st problem: Transient analysis (without Damping)

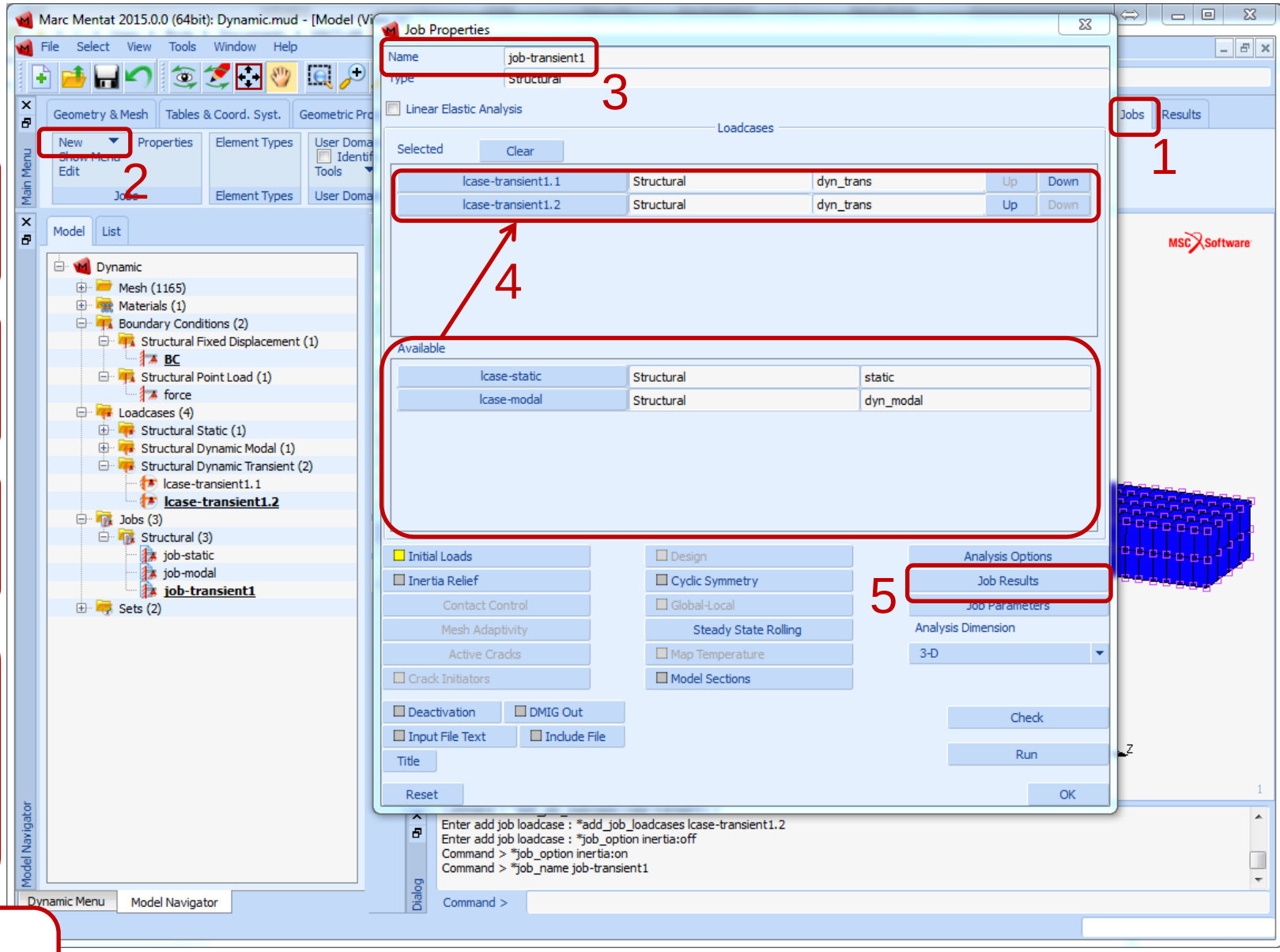
Step 1:
Go to *Job* Tab from *Main Menu*.

Step 2:
Click on *New* list and choose *Structural*.

Step 3:
Change *Name* to „job-transient1”.

Step 4:
Select:
*lcase-transient1.1* and *lcase-transient1.2* loadcases from *Available* Area.

Step 5:
Click *Job Results* Button to open *Job Results* Window.






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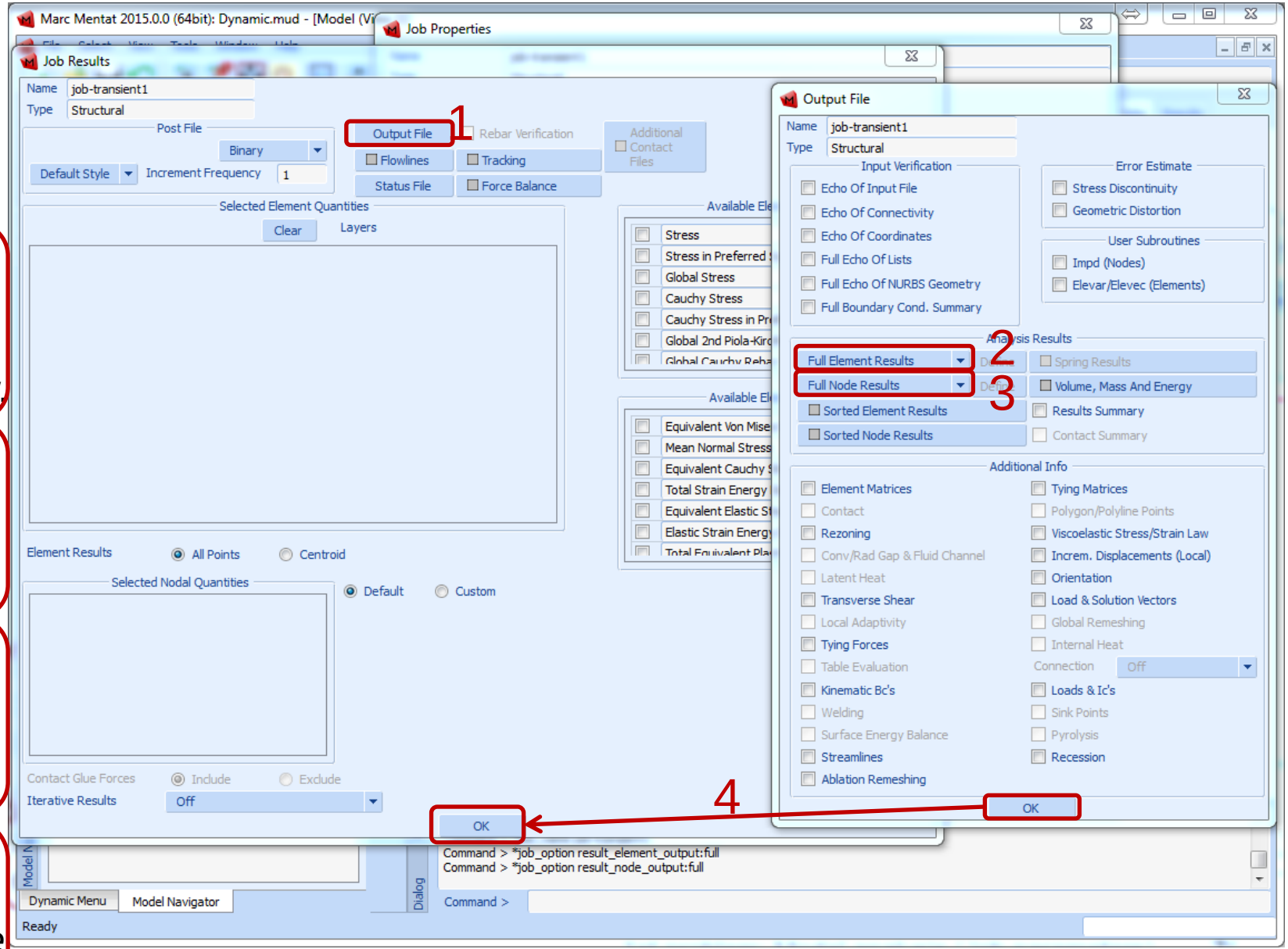
# 1st problem: Transient analysis (without Damping)

Step 1:
Click *Output File* Button to open *Output File* Window.

Step 2:
Choose *Full Element Results* Option from the list.

Step 3:
Choose *Full Node Results* Option from the list.

Step 4:
Click *OK* Buttons to close both *Output File* and *Job Results* Windows.






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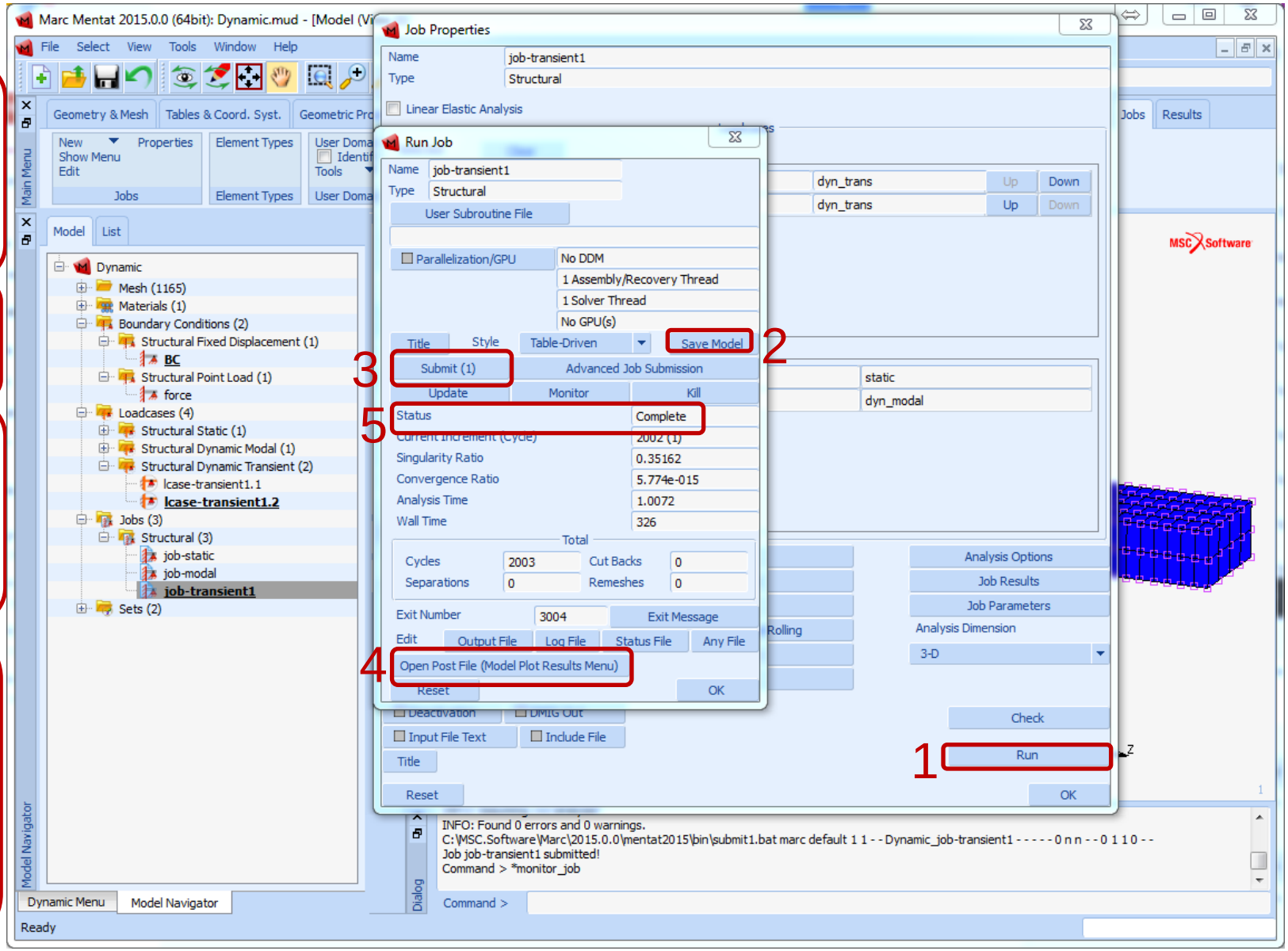
# 1st problem: Transient analysis (without Damping)

Step 1:
Click *Run* Button to open *Run Job* Window.

Step 2:
Save model.

Step 3:
Run analysis by clicking *Submit(1)* Button.

Step 4:
Click *Open Post File…* Button after *Status* Bar change to *Complete* 5 .






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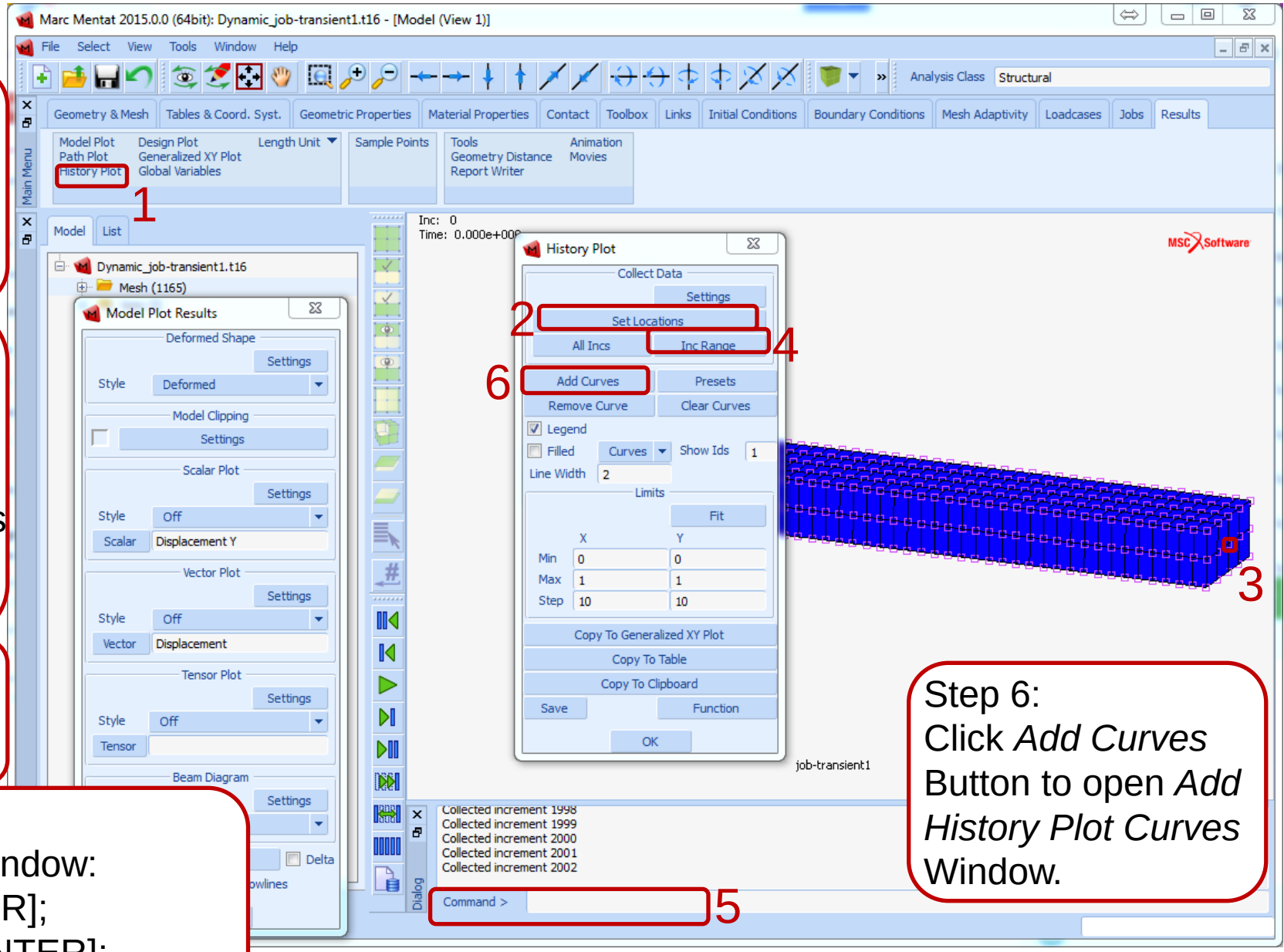
# 1st problem: Transient analysis (without Damping)

Step 1:
Click *History Plot* Button to open *History Plot* Window.

Step 2:
Click *Set Locations* Button and pick (click LMB) the node where force is applied 3. Accept choice (click RBM)

Step 4:
Click *Inc Range* Button.

Step 5:
Input to Command Window:
-> 1 (first step) [ENTER];
-> 2002 (last step) [ENTER];
-> 1 (increment step size) [ENTER].




Step 6:
Click *Add Curves* Button to open *Add History Plot Curves* Window.



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# 1st problem: Transient analysis (without Damping)

Step: 1
Click *Single Location* Button.

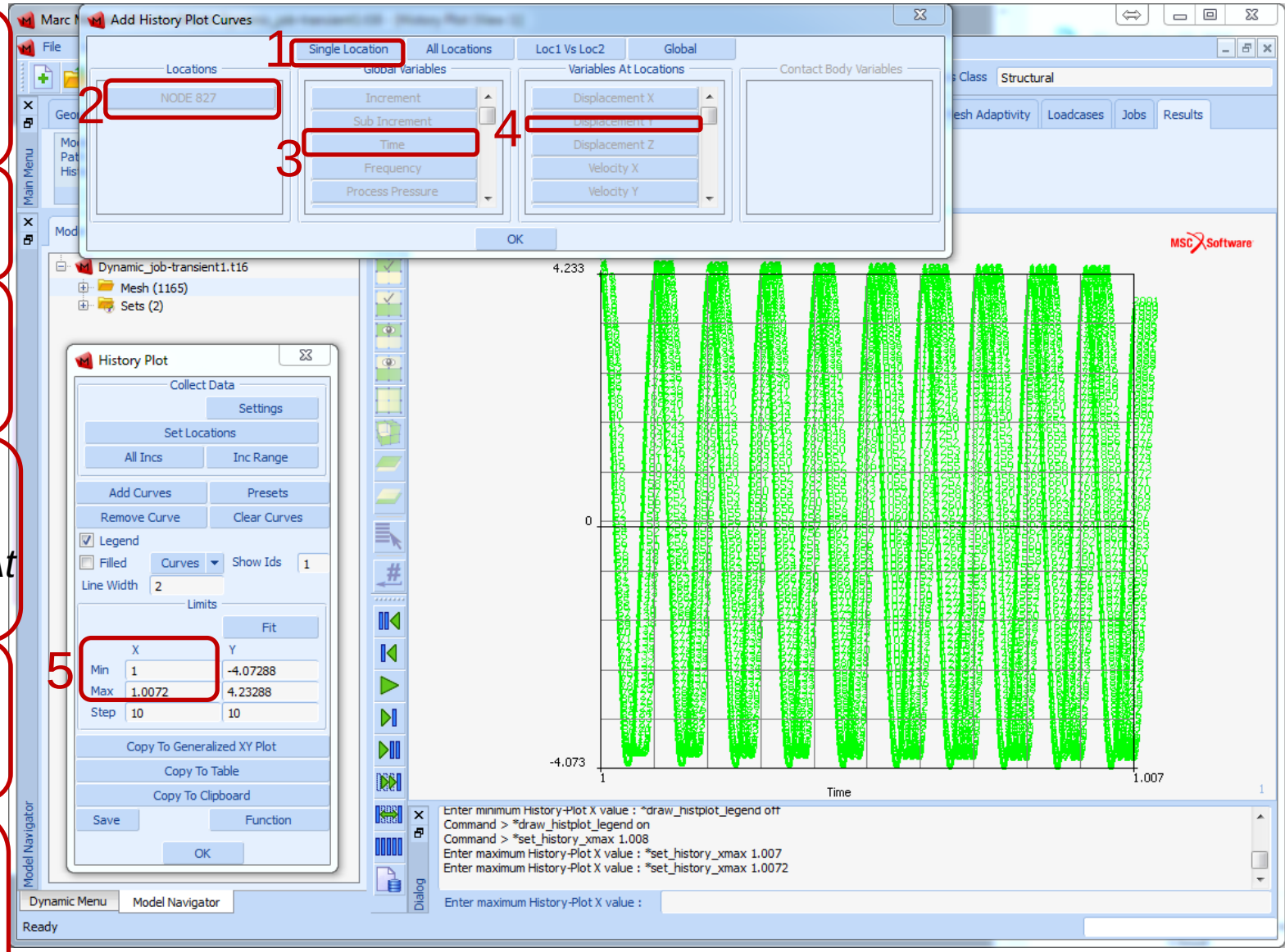
Step 2:
Click *Node 827* Button.

Step 3:
Click *Time* Button from *Global Variables* Area.

Step 4:
Click *Displacement Y* Button from *Variables At Locations* Area.

Step 5:
Change range of X axis to <1, 1.0072>.

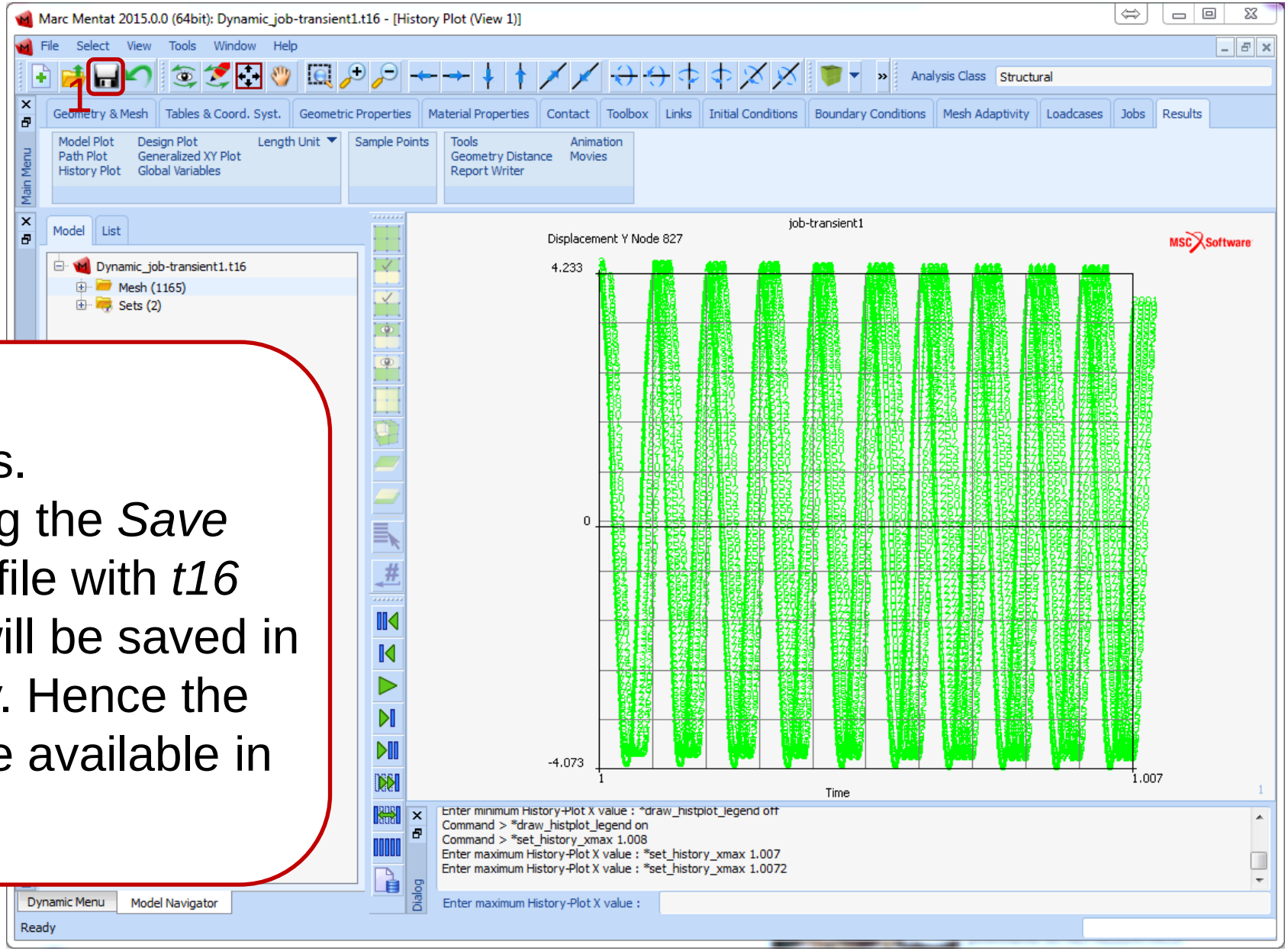
One can observe **almost** 10 cycles of the beam deflection because calculated time is rounded down.






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# 1st problem: Transient analysis (without Damping)




Step 1:
Save results.
After clicking the *Save* Button, the file with *t16* extension will be saved in PC memory. Hence the result will be available in future.



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# 1st problem: Transient analysis (with Damping)

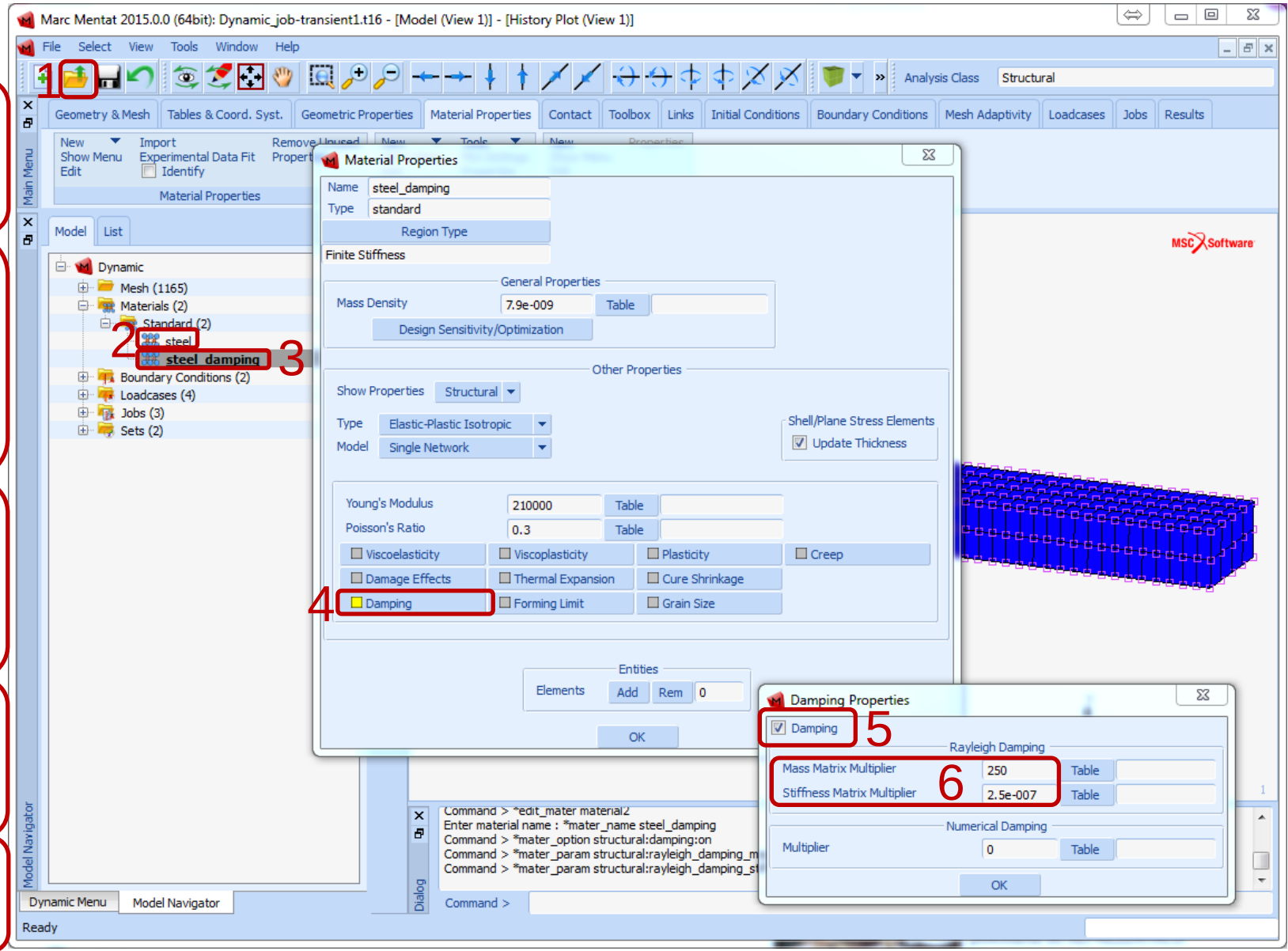
Step 1:
Open model with the Beam.

Step 2:
Click RBM on *steel* icon located at Tree and choose *Copy* Option.

Step 3:
Open Material Properties Windows by double clicking.

Step 4:
Click *Damping* Button and activate option 5.

Step 6:
Input damping data.



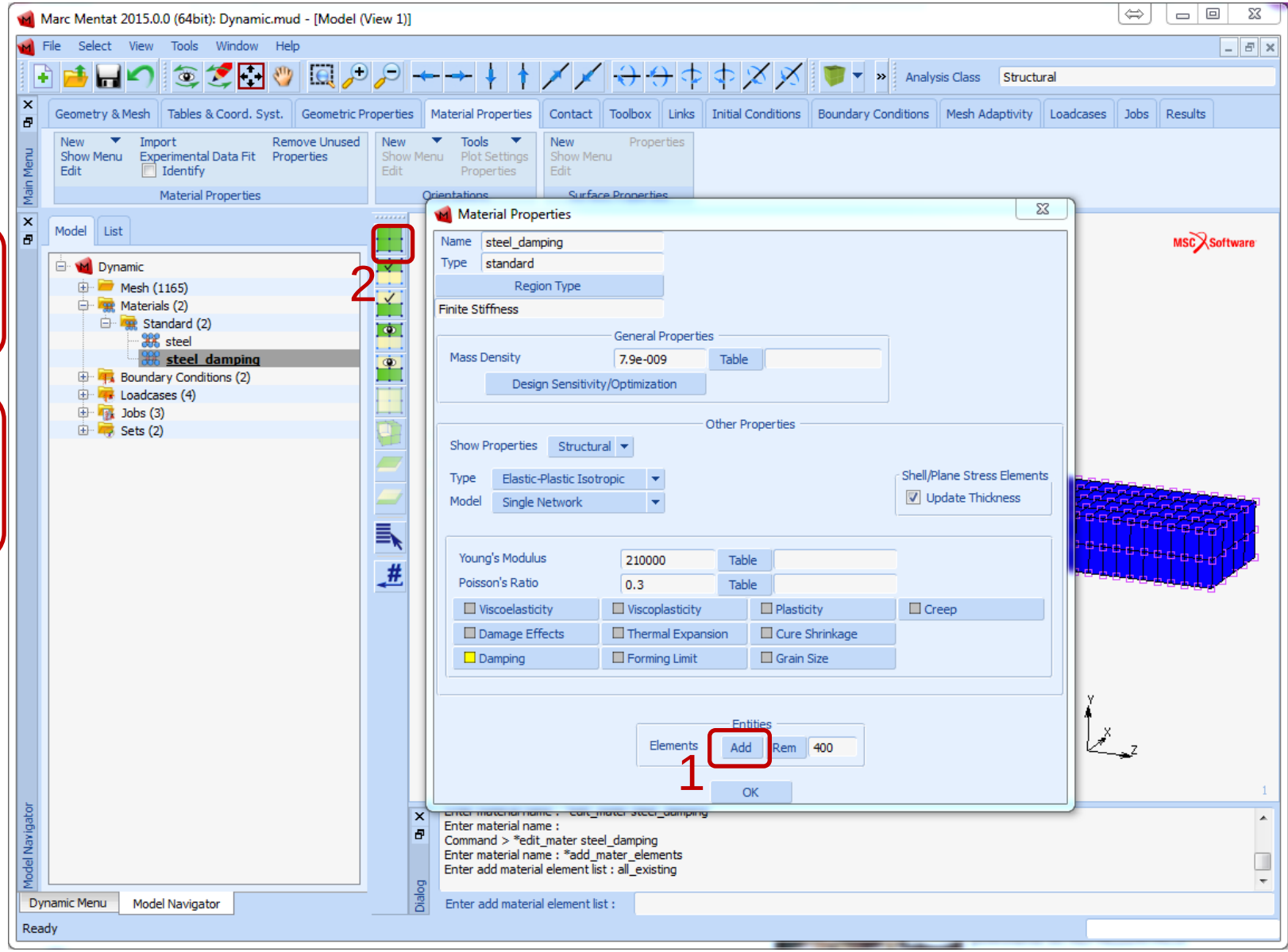



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# 1st problem: Transient analysis (with Damping)

Step 1:
Click *Add* Button.

Step 2:
Click *all_exisiting* Button.






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# 1st problem: Transient analysis (with Damping)

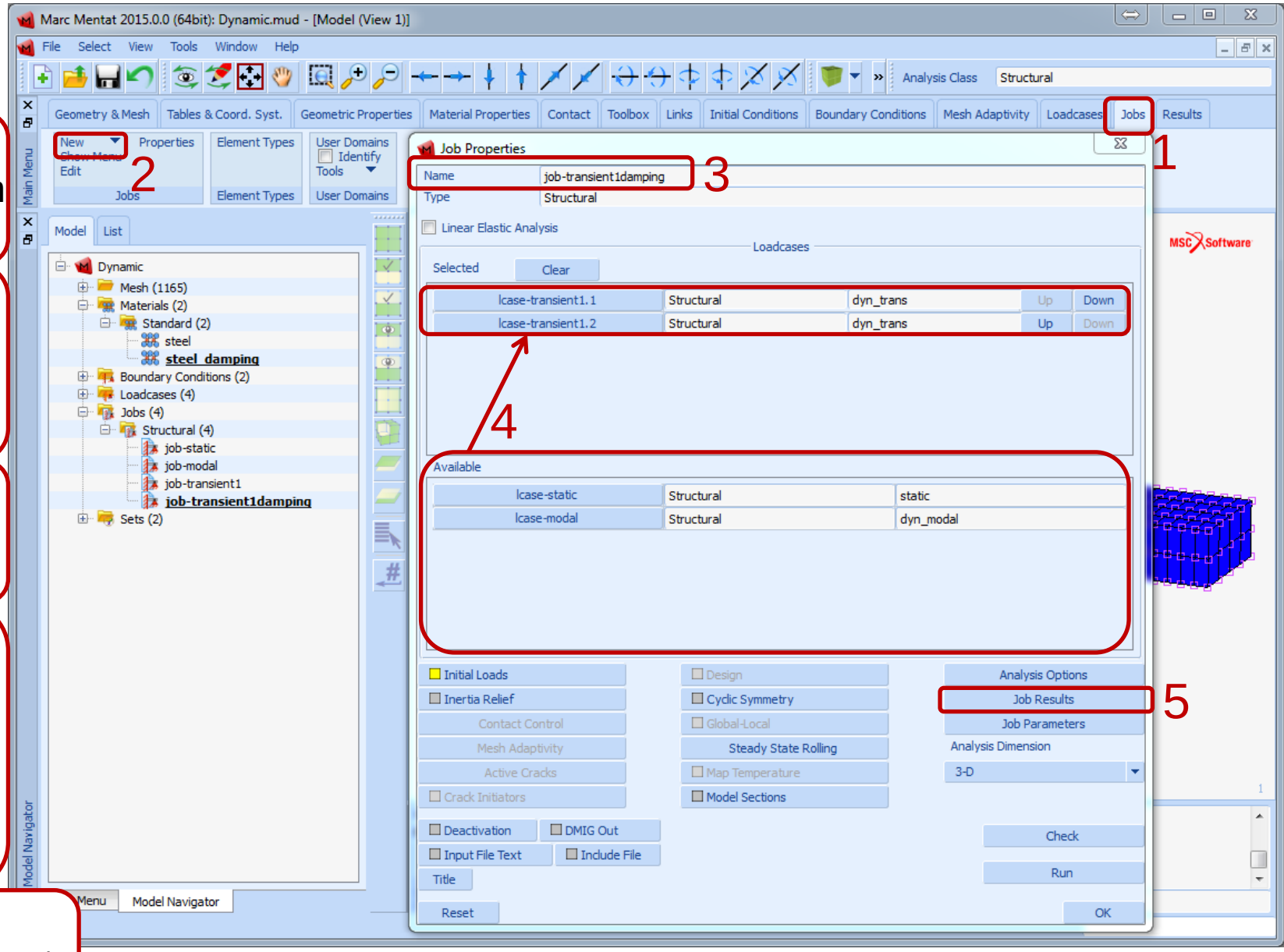
Step 1:
Go to *Jobs* Tab from *Main Menu*.

Step 2:
Click *New* list and choose *Structural* Option.

Step 3:
Change Name to „*job-transient1damping*”.

Step 4:
Select:
*lcase-transient1.1* and *lcase-transient1.2* loadcases from *Available* Area.

Step 5:
Click *Job Results* Button to open *Job Results* Window.



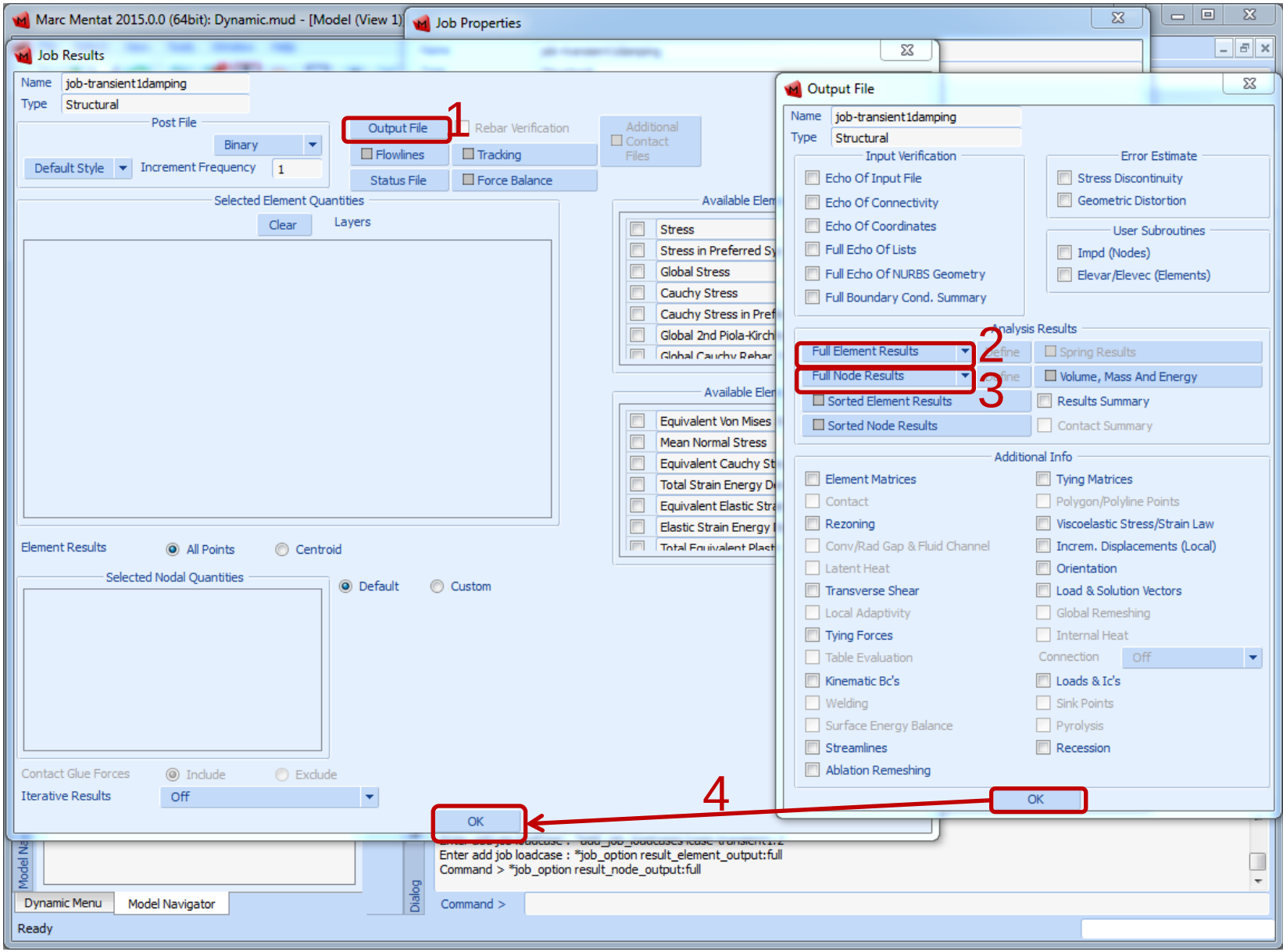
# 1st problem: Transient analysis (without Damping)

Step 1:
Click *Output File* Button to open *Output File* Window.

Step 2:
Choose *Full Element Results* Option from the list.

Step 3:
Choose *Full Node Results* Option from the list.

Step 4:
Click *OK* Buttons to close both *Output File* and *Job Results* Windows.






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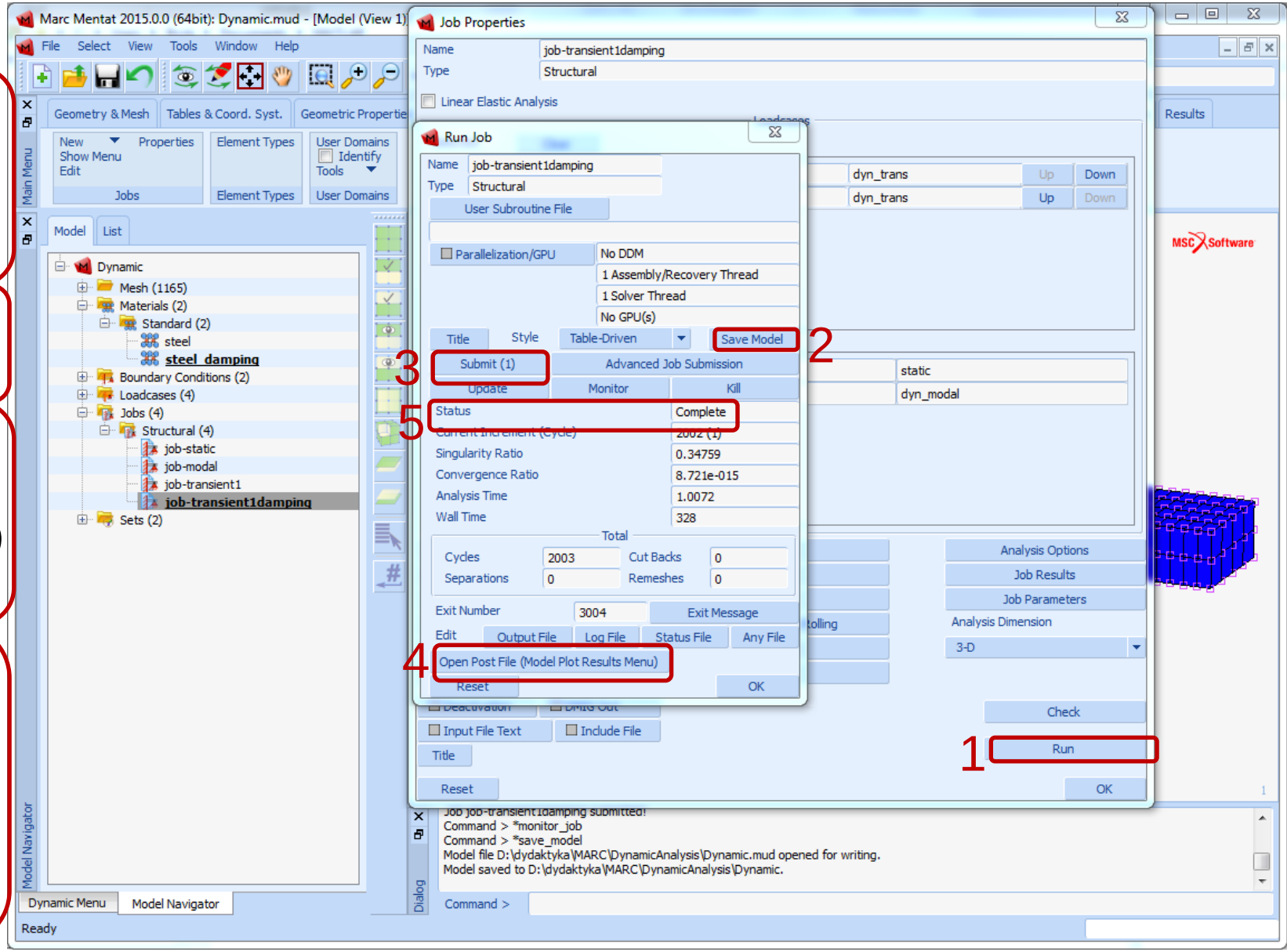
# 1st problem: Transient analysis (with Damping)

Step 1:
Click *Run* Button to open *Run Job* Window.

Step 2:
Save model.

Step 3:
Run analysis by clicking *Submit(1)* Button.

Step 4:
Click *Open Post File…* Button after *Status* Bar change to *Complete* 5 .






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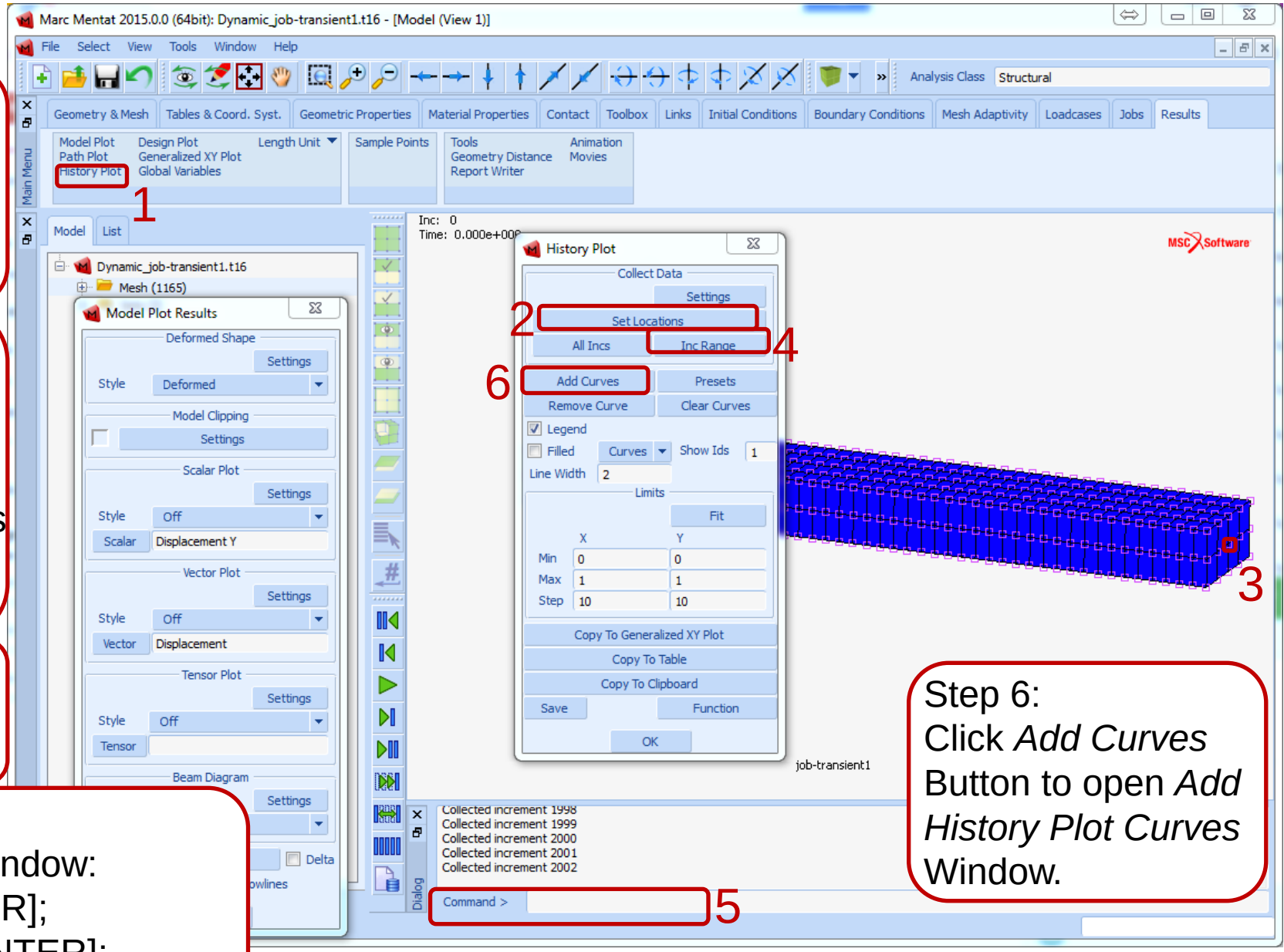
# 1st problem: Transient analysis (with Damping)

Step 1:
Click *History Plot* Button to open *History Plot* Window.

Step 2:
Click *Set Locations* Button and pick (click LMB) the node where force is applied 3 . Accept choice (click RBM)

Step 4:
Click *Inc Range* Button.

Step 5:
Input to Command Window:
-> 1 (first step) [ENTER];
-> 2002 (last step) [ENTER];
-> 1 (increment step size) [ENTER].




Step 6:
Click *Add Curves* Button to open *Add History Plot Curves* Window.



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# 1st problem: Transient analysis (with Damping)

Step: 1
Click *Single Location* Button.

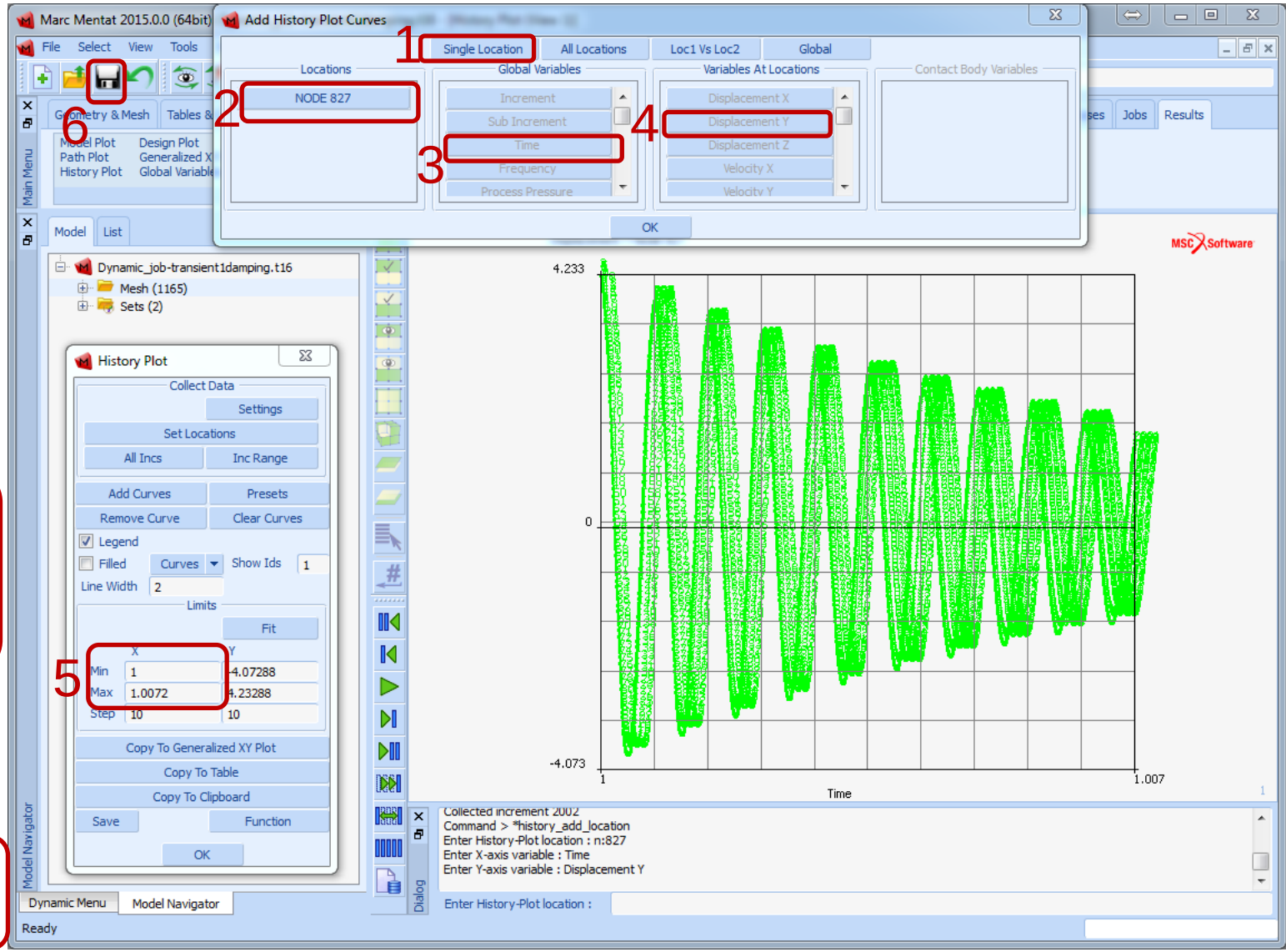
Step 2:
Click *Node 827* Button.

Step 3:
Click *Time* Button from *Global Variables* Area.

Step 4:
Click *Displacement Y* Button from *Variables At Locations* Area.

Step 5:
Change range of X axis to <1, 1.0072>.

Step 6:
Save results to file.






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# 2nd problem: Harmonic analysis

Step 1:
Open model with the Beam.

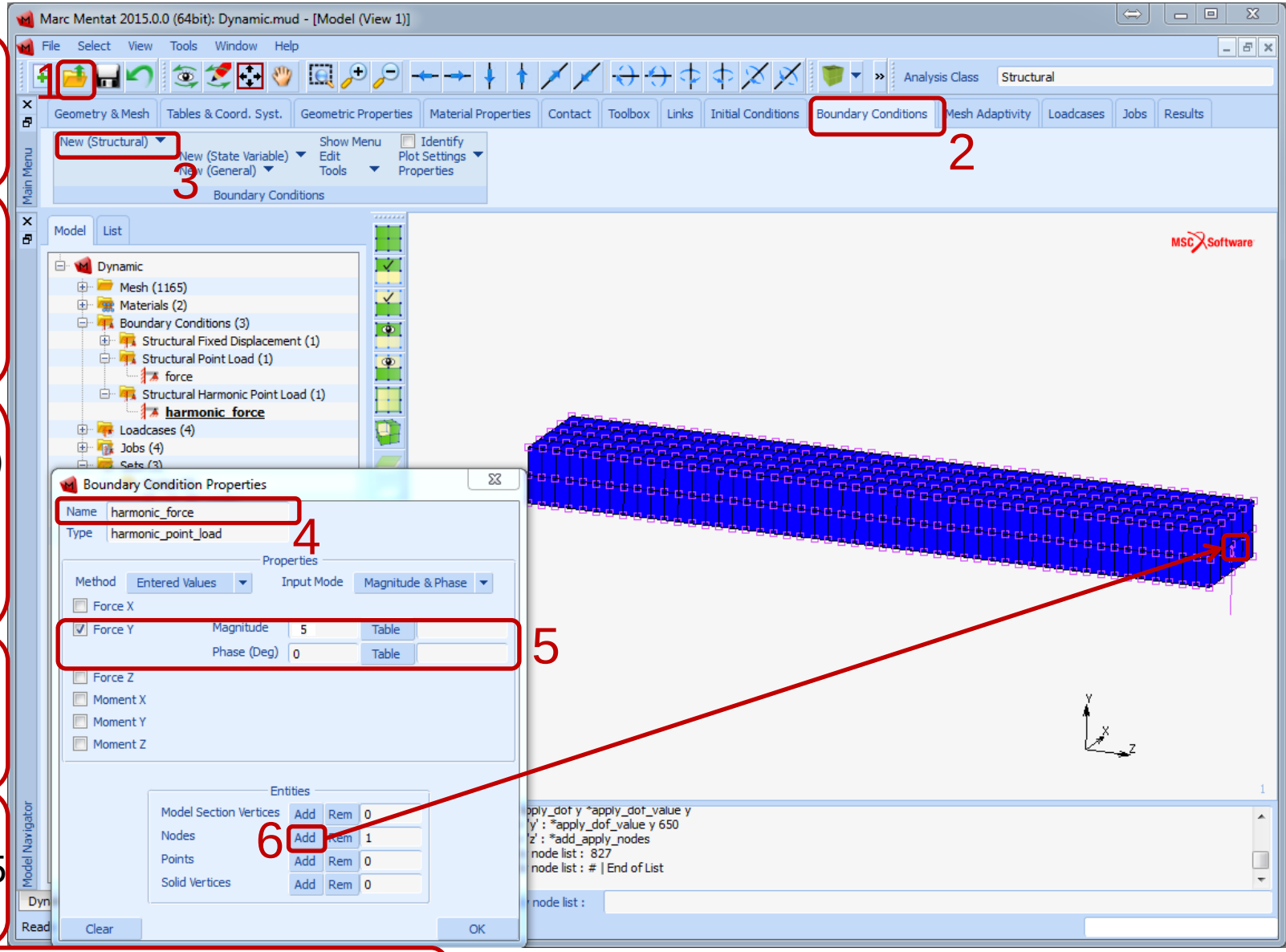
Step 2:
Go to *Boundary Conditions* Tab from *Main Menu*.

Step 3:
Click *New (Structural)* list and choose *Harmonic Point Load* Option.

Step 4:
Change *Name* to „harmonic_load”.

Step 5:
Pick *Force Y* and input 5 as *Magnitude* value.

Step 6:
Click Add Button and choose node with index 827.





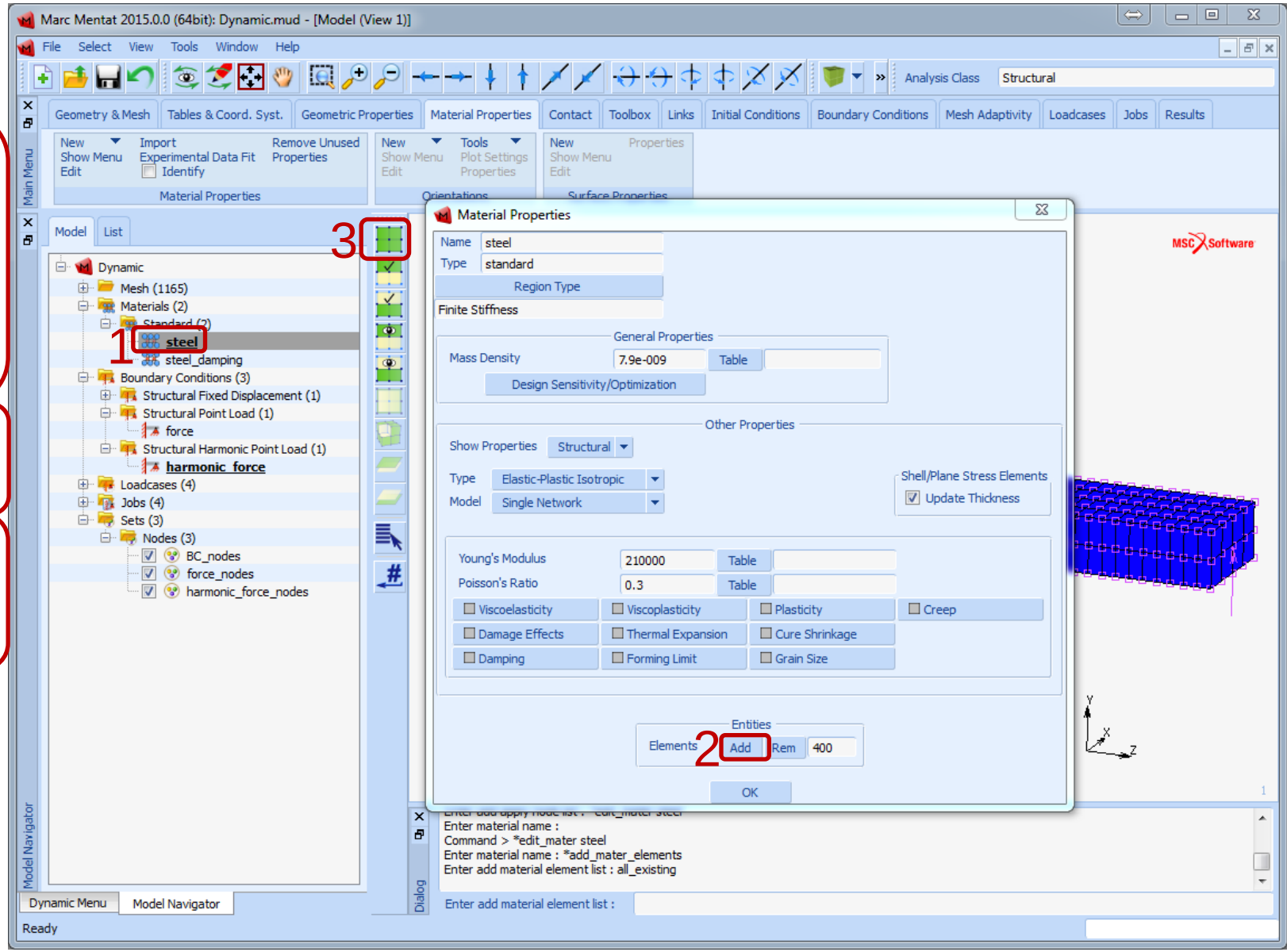

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# 2nd problem: Harmonic analysis

Step 1:
Double click on *steel* icon located at *Tree* to open *Material Properties* Window.

Step 2:
Click *Add* Button.

Step 3:
Click *all_existing* Button.






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# 2nd problem: Harmonic analysis

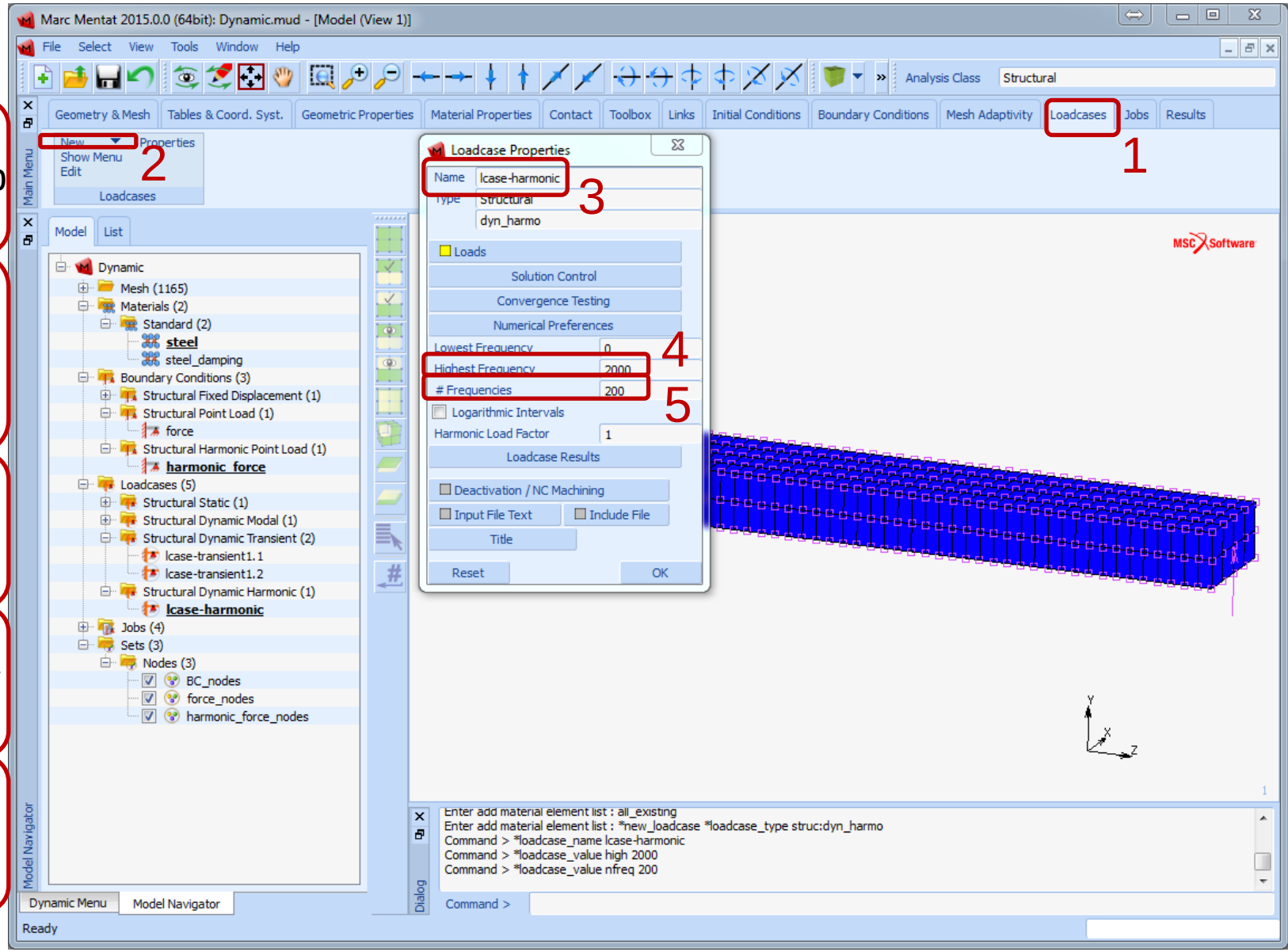
Step 1:
Go to *Loadcases* Tab from *Main Menu*.

Step 2:
Click *New* list and choose *Dynamic Harmonic* Option.

Step 3:
Change *Name* to „lcase-harmonic”.

Step 4:
Set *Highest Frequency* to 2000 [Hz].

Step 5:
Set # *Frequencies* to 200.



# 2nd problem: Harmonic analysis

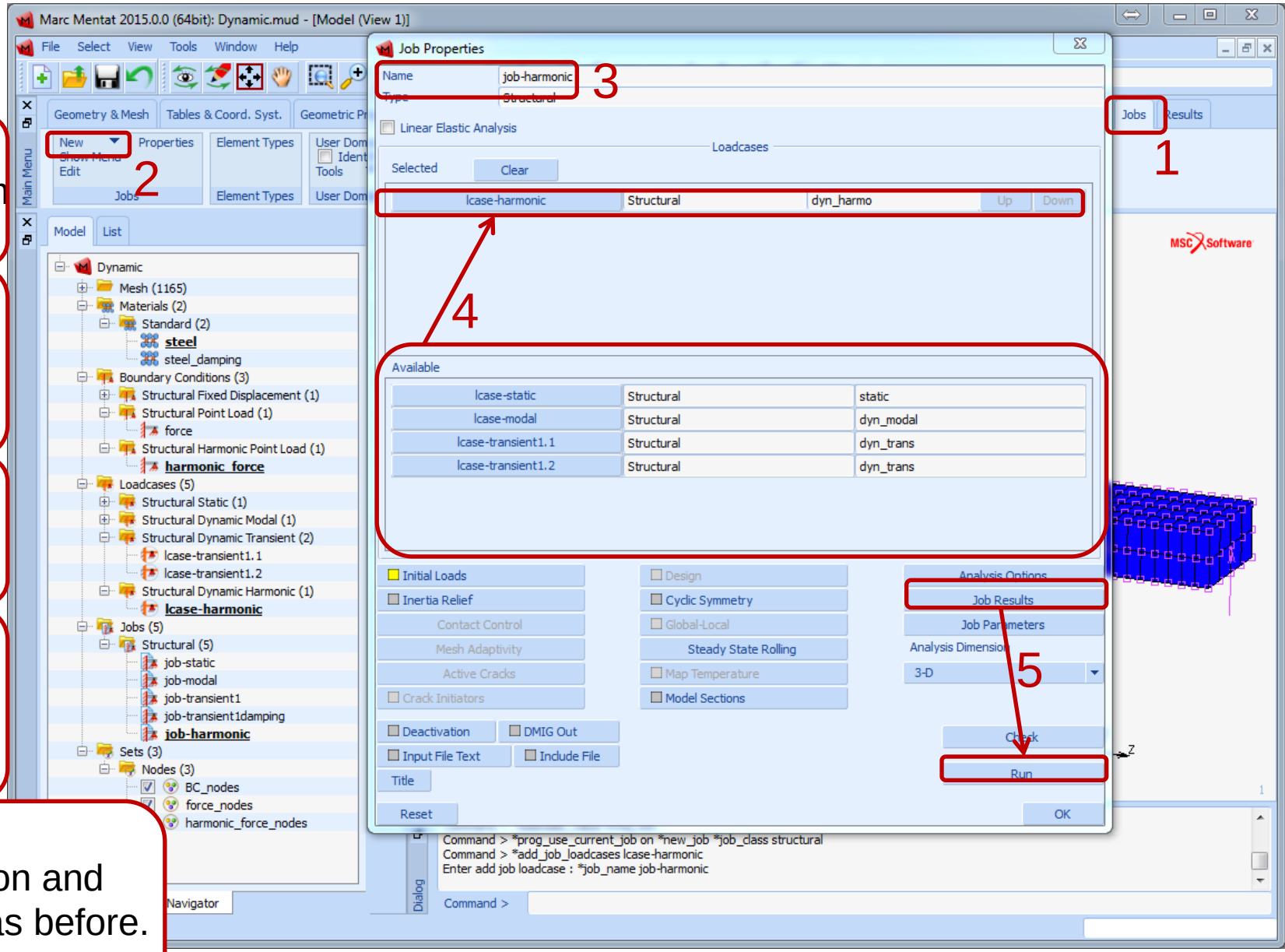
Step 1:
Go to *Jobs* Tab from *Main Menu*.

Step 2:
Click *New* list and choose *Structural* Option.

Step 3:
Change *Name* to „job-harmonic”.

Step 4:
Select *lcase-harmonic* loadcase from *Available* Area.

Step 5:
Click *Job Results* Button and configure Output File as before. Next, Run simulation and open Model Plot Results Menu.



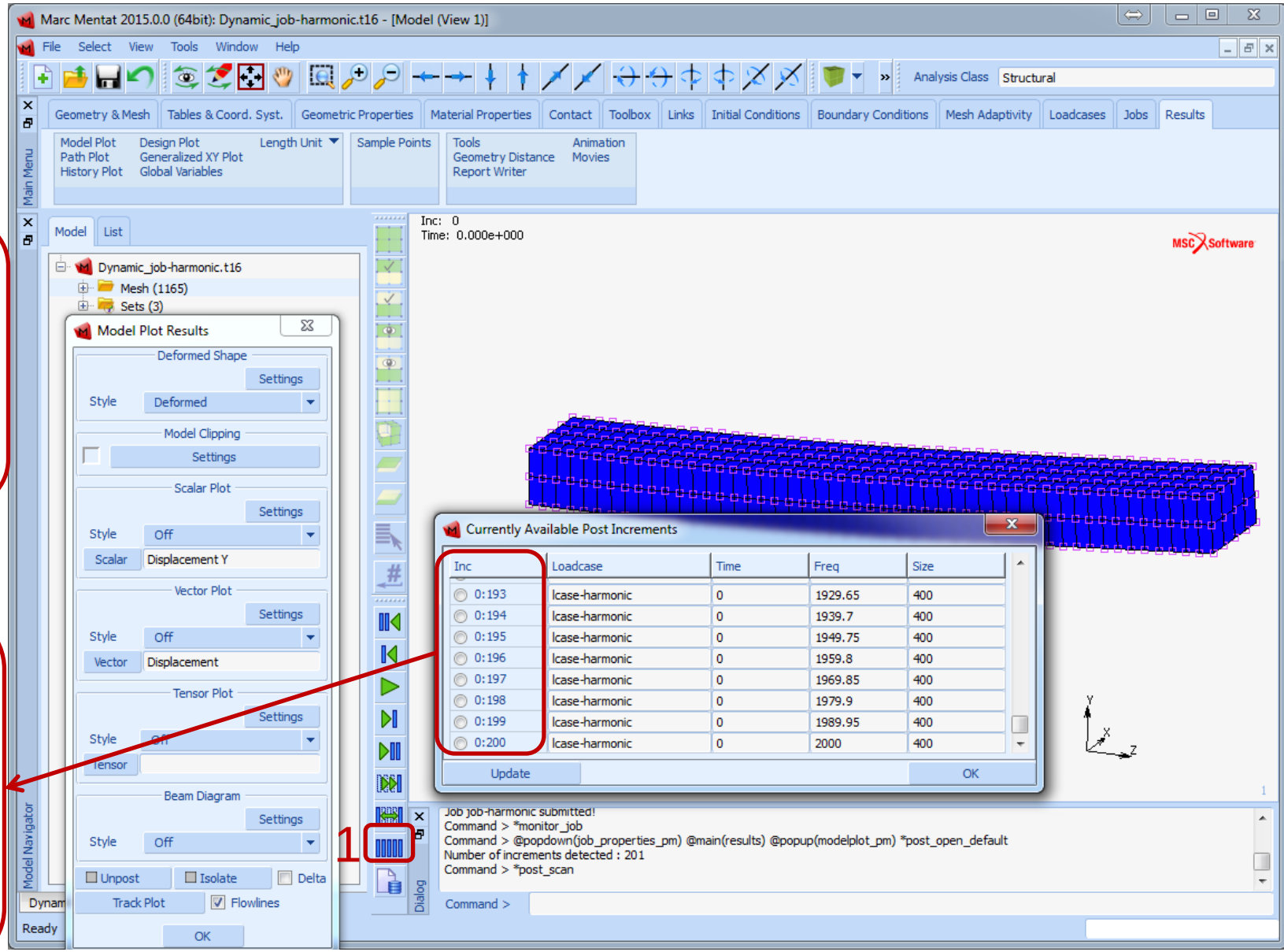



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# 2nd problem: Harmonic analysis

Step 1:
Click *Scan results file* Button to open *Currently Available Post Increments* Window.

Observe that in case of harmonic analysis the steps (Inc) have different names than in transient analysis. It is important according to step 2 in next slide.






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# 2nd problem: Harmonic analysis

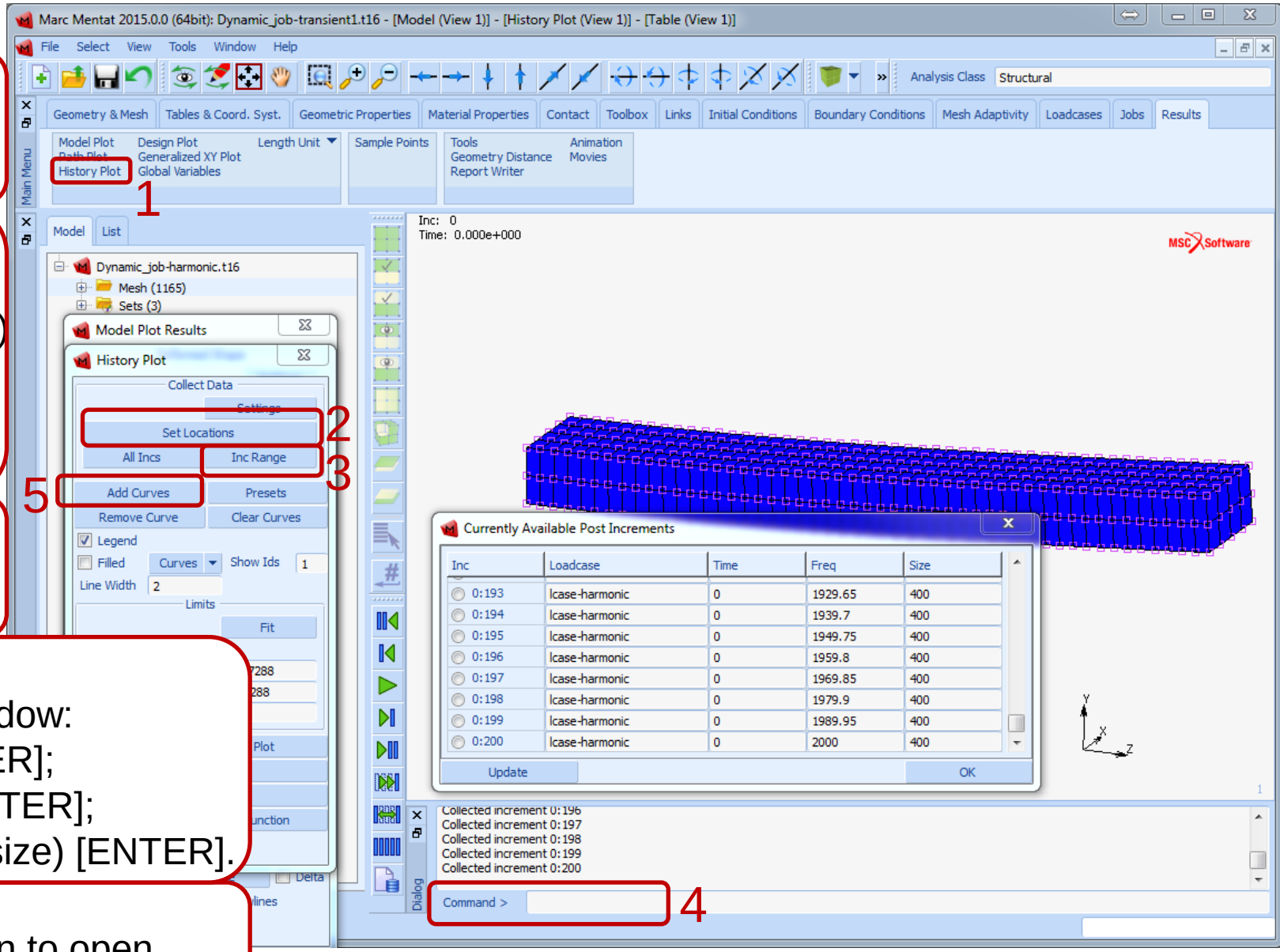
Step 1:
Click *History Plot* Button.

Step 2:
Click *Set Location* Button, choose (LBM) node with index 827 and accept choice (RBM).

Step 3:
Click *Inc Range* Button.

Step 4:
Input to Command Window:
-> 0:1 (first step) [ENTER];
-> 0:200 (last step) [ENTER];
-> 1 (incremental step size) [ENTER].

Step 5:
Click Add Curves Button to open Add History Plot Curves Window.






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# 2nd problem: Harmonic analysis

Step 1:
Click *Single Location* Button.

Step 2:
Click NODE827 Button.

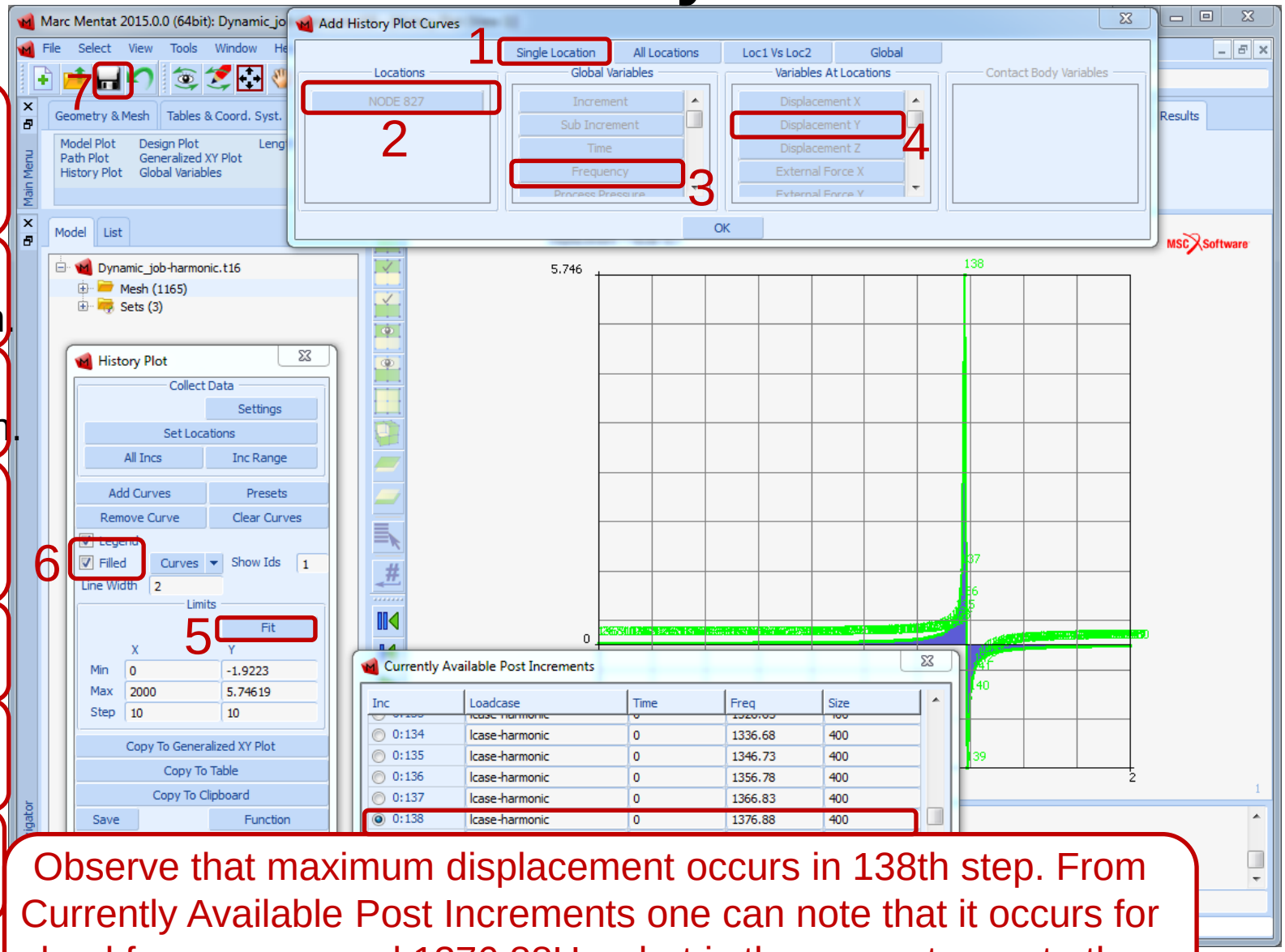
Step 3:
Click *Frequency* Button.

Step 4:
Click *Displacement Y* Button.

Step 5:
Click *Fit* Button.

Step 6:
Pick Filled Option.

Step 7:
Save results.




Observe that maximum displacement occurs in 138th step. From Currently Available Post Increments one can note that it occurs for load frequency equal 1376.88Hz what is the nearest case to the beam modal frequency (1379.41Hz).

# 2nd problem: Transient analysis

Step 1:
Open model with the Beam.

Step 2 & 3:
Go to *Tables & Coord. Syst.* Tab from *Main Menu*, click *New* and choose first Option.

Step 4:
Change *Name* to „sin1379.41”.

Step 5:
Click *Type* Button to open *Independent Variable Types* Window.

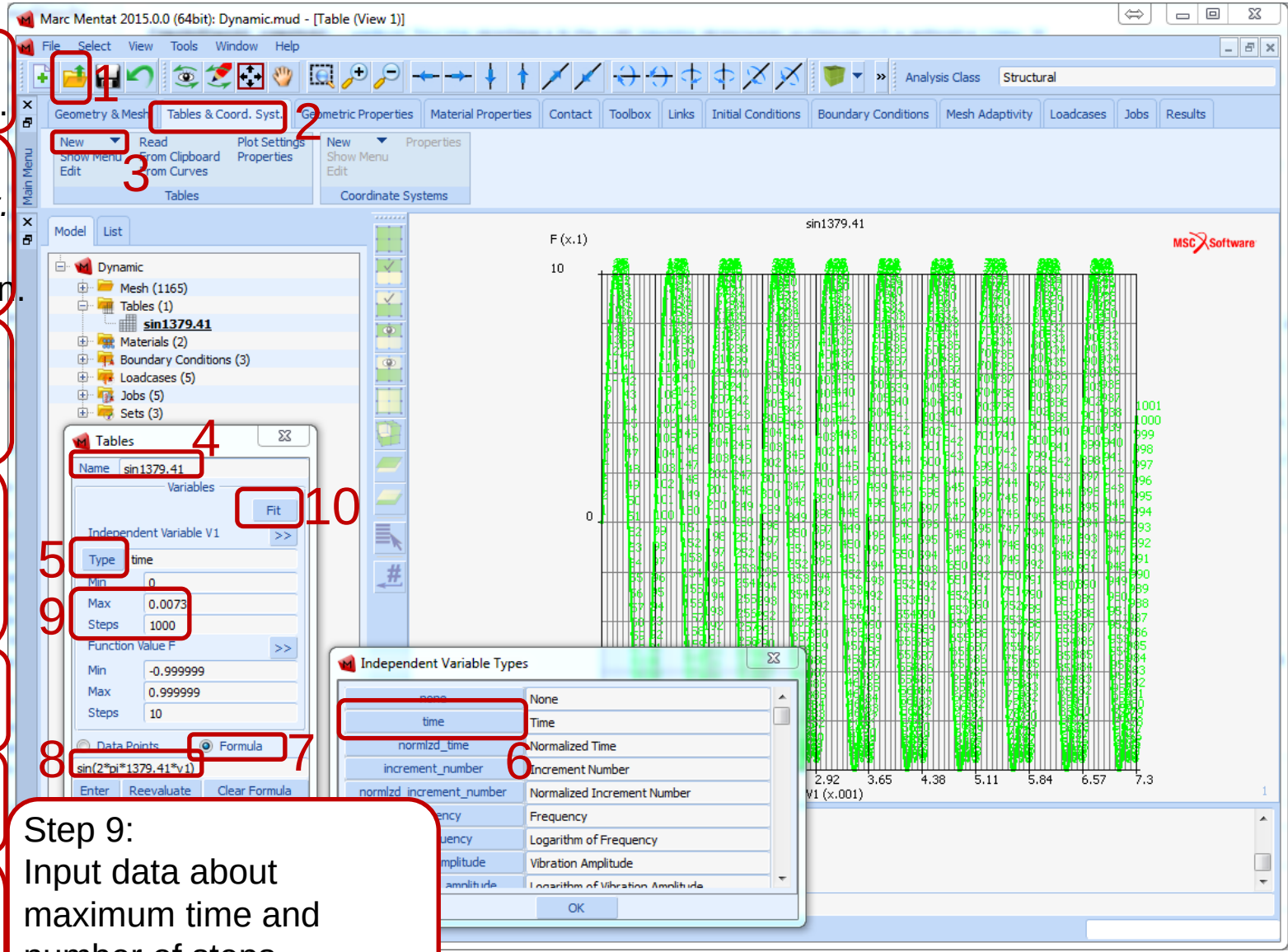
Step 6:
Click *time* Button.

Step 7:
Pick *Formula* Option.

Step 8:
Input Formula:
sin(2*pi*1379.41*v1). V1 is intependent variable – in this case: time.

Step 9:
Input data about maximum time and number of steps.

Step 10:
Click *Fit* Button.






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# 2nd problem: Transient analysis

Step 1:
Click *Window* list and choose *Model (View1)* Option.

Step 2:
Go to *Boundaty Conditions* Tab from *Main Menu*.

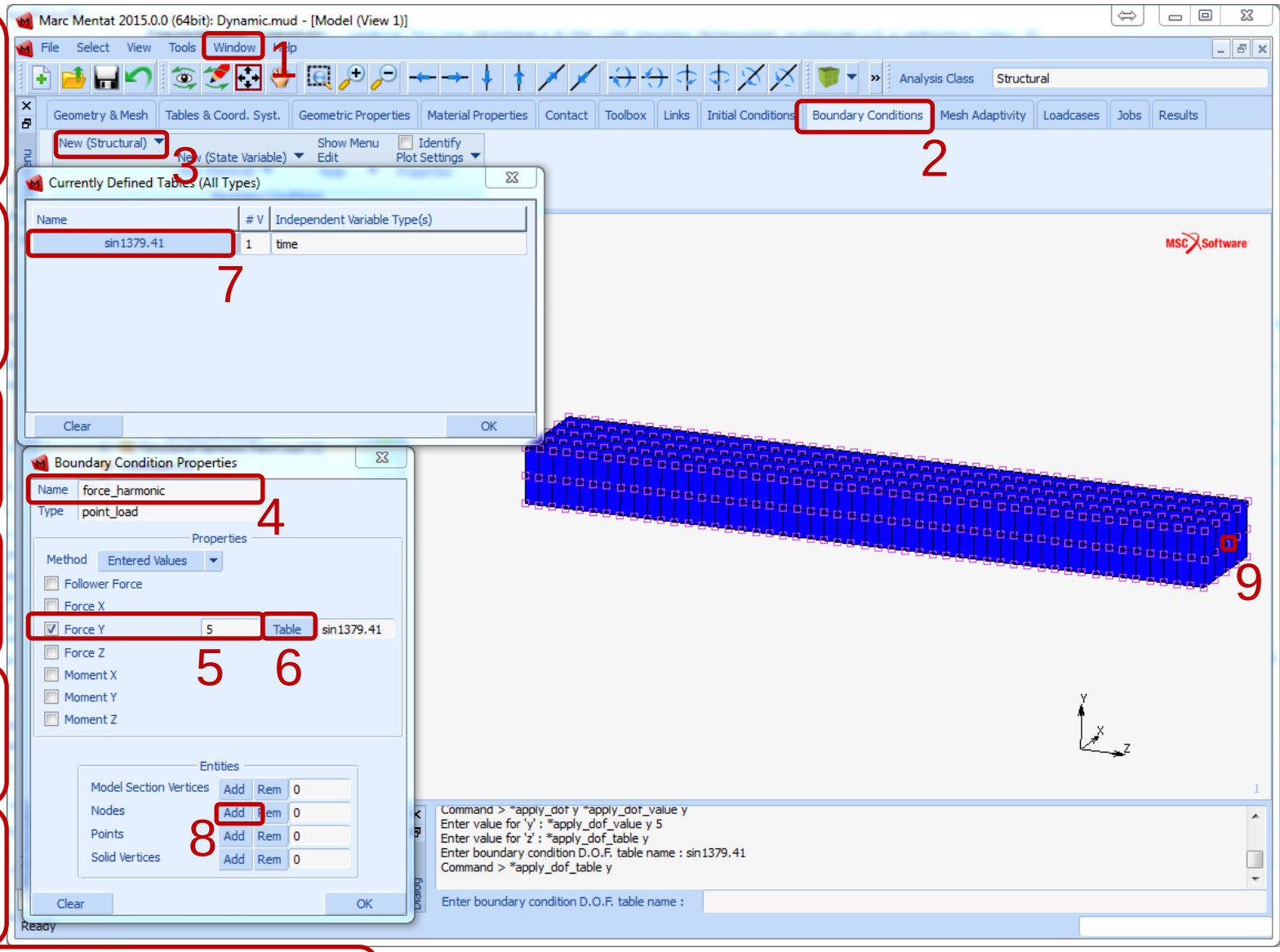
Step 3:
Click *New* list and choose *Point Load* Option.

Step 4:
Change *Name* to „force_harmonic”.

Step 5:
Pick Force Y Option and apply 5 [N].

Step 6:
Click *Table* Button and next *sin1379.41* Button 7.

Step 8:
Click *Add* Button and choose node with index 827 9






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# 2nd problem: Transient analysis

Step 1:
Go to *Loadcases* Tab from *Main Manu*.

Step 2:
Click *New* list and choose *Dynamic Transient* Option.

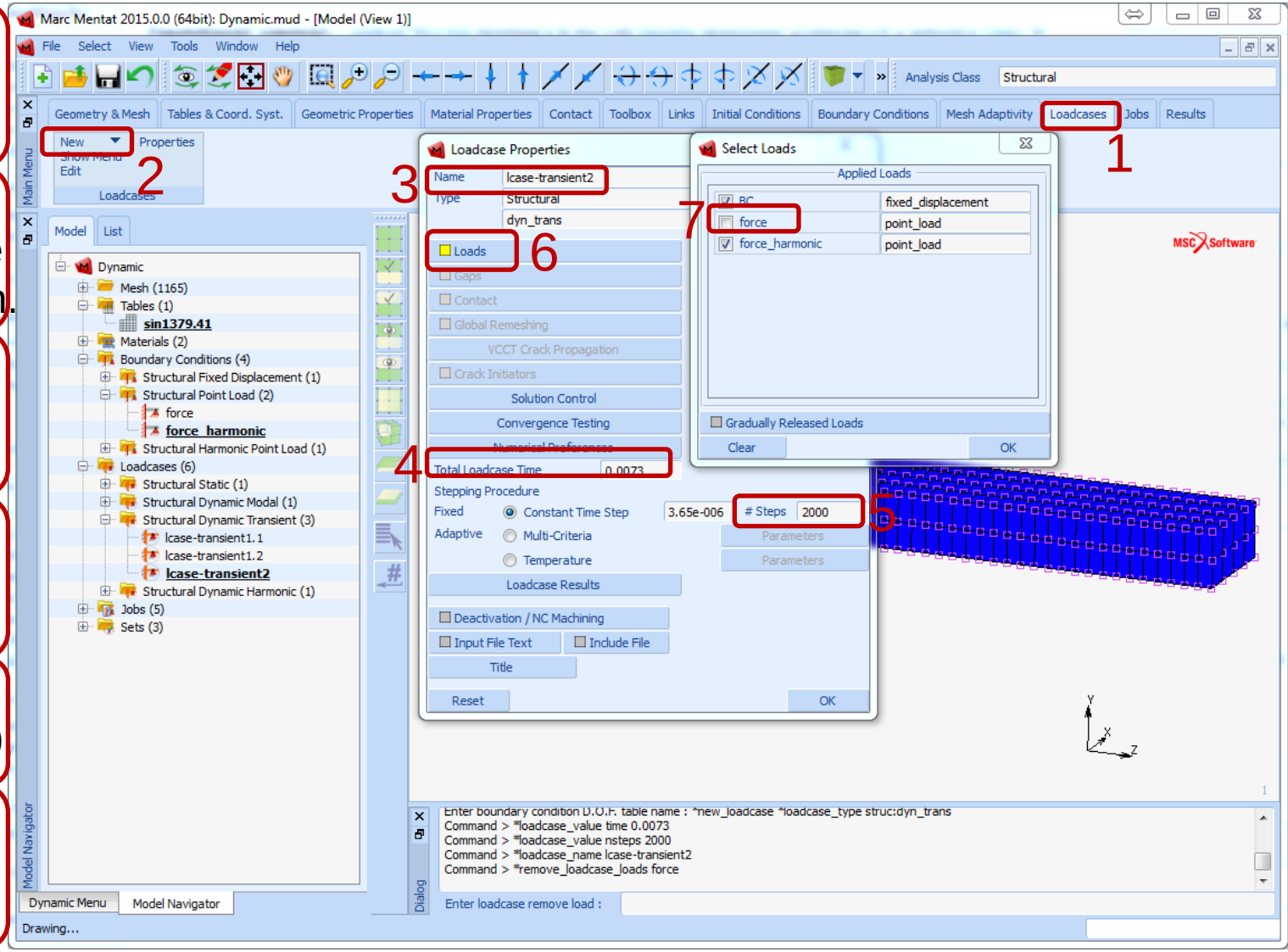
Step 3:
Change *Name* to „lcase-transient2”.

Step 4:
Set *Total Loadcase Time* to 0.0073.

Step 5:
Set number of *Steps* to 2000

Step 6:
Click *Loads* Button to open *Select Loads* Window.

Step 7:
Unselect *force*.






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# 2nd problem: Transient analysis

Step 1:
Go to *Jobs* Tab from *Main Menu*.

Step 2:
Click *New* list and choose *Structural* Option.

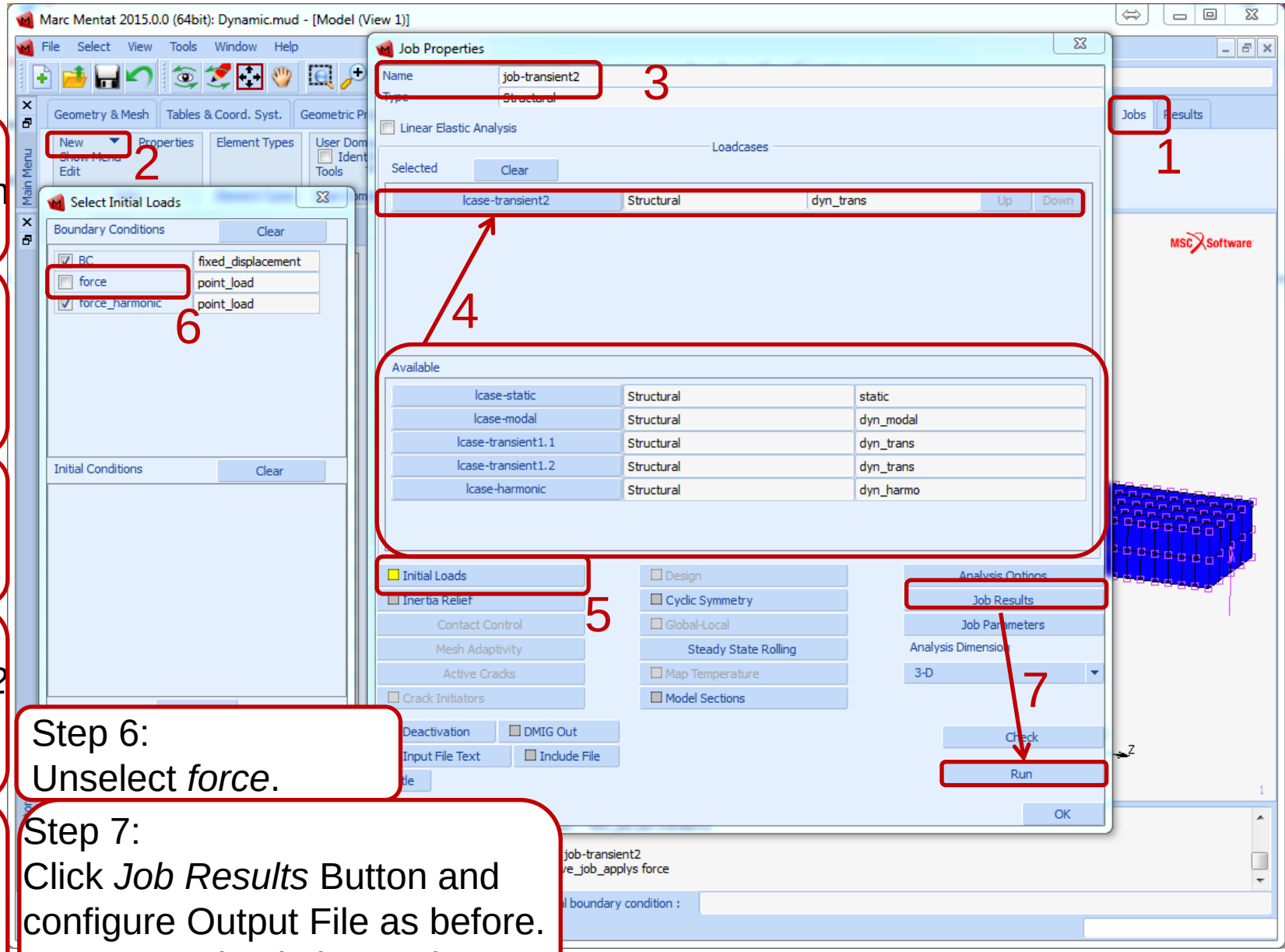
Step 3:
Change *Name* to „job-transient2”.

Step 4:
Select *lcase-transient2* loadcase from *Available* Area.

Step 5:
Click *Initial Loads* Button to open *Select Initial Loads* Window.

Step 6:
Unselect *force*.

Step 7:
Click *Job Results* Button and configure Output File as before. Next, Run simulation and open Model Plot Results Menu.






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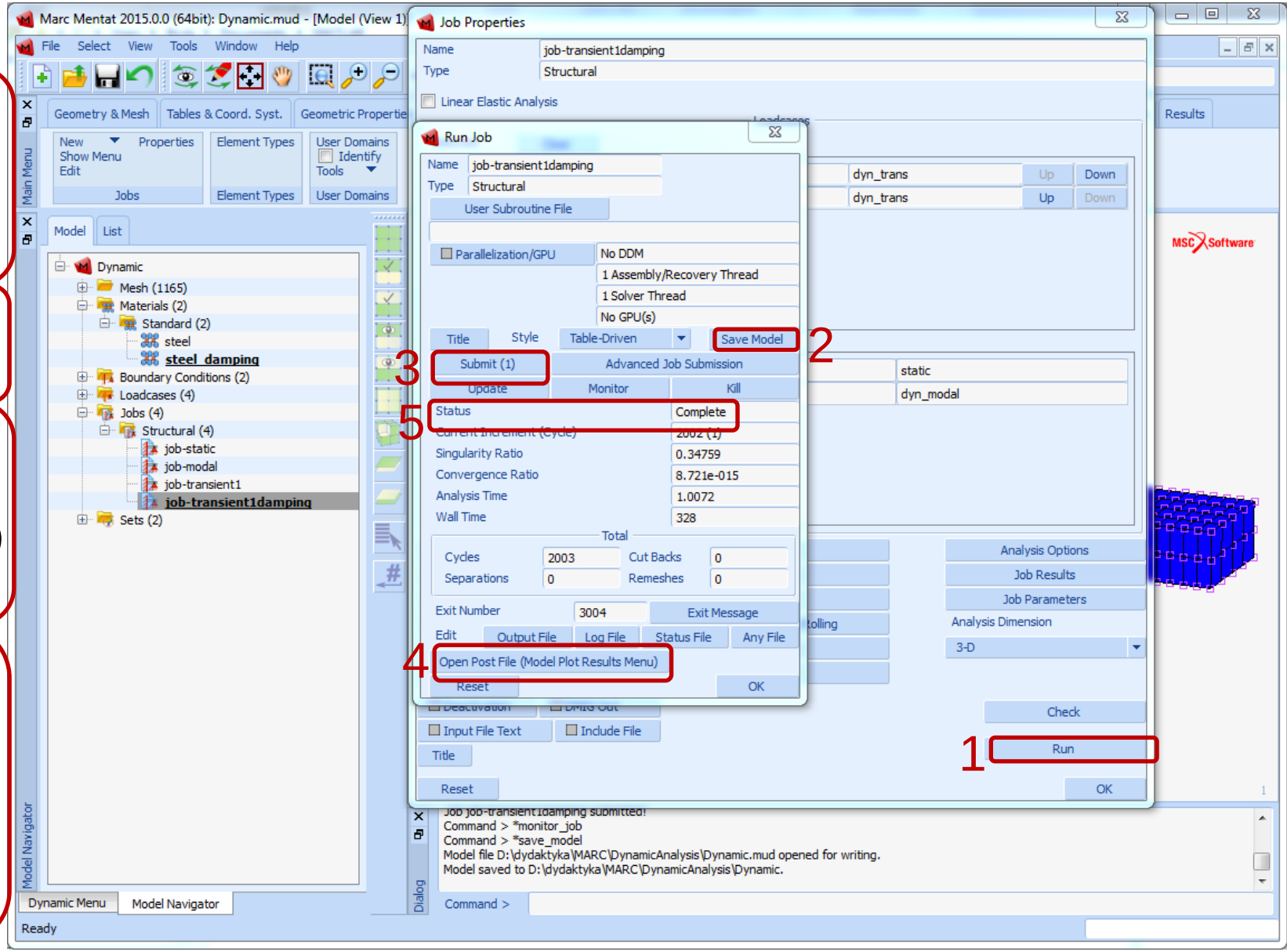
# 2nd problem: Transient analysis

Step 1:
Click *Run* Button to open *Run Job* Window.

Step 2:
Save model.

Step 3:
Run analysis by clicking *Submit(1)* Button.

Step 4:
Click *Open Post File…* Button after *Status* Bar change to *Complete* 5 .



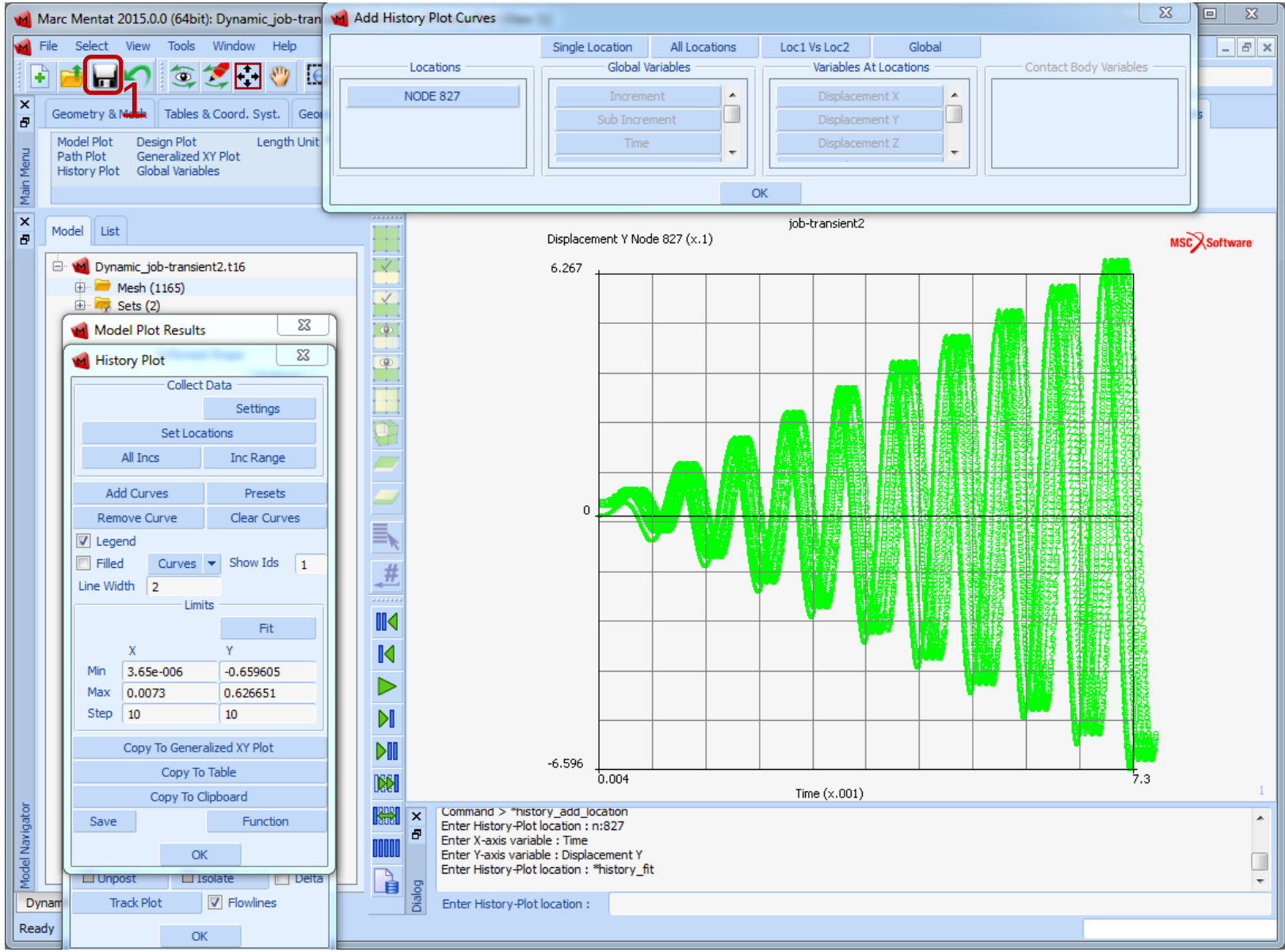



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# 2nd problem: Transient analysis

Step 1:
Save results.

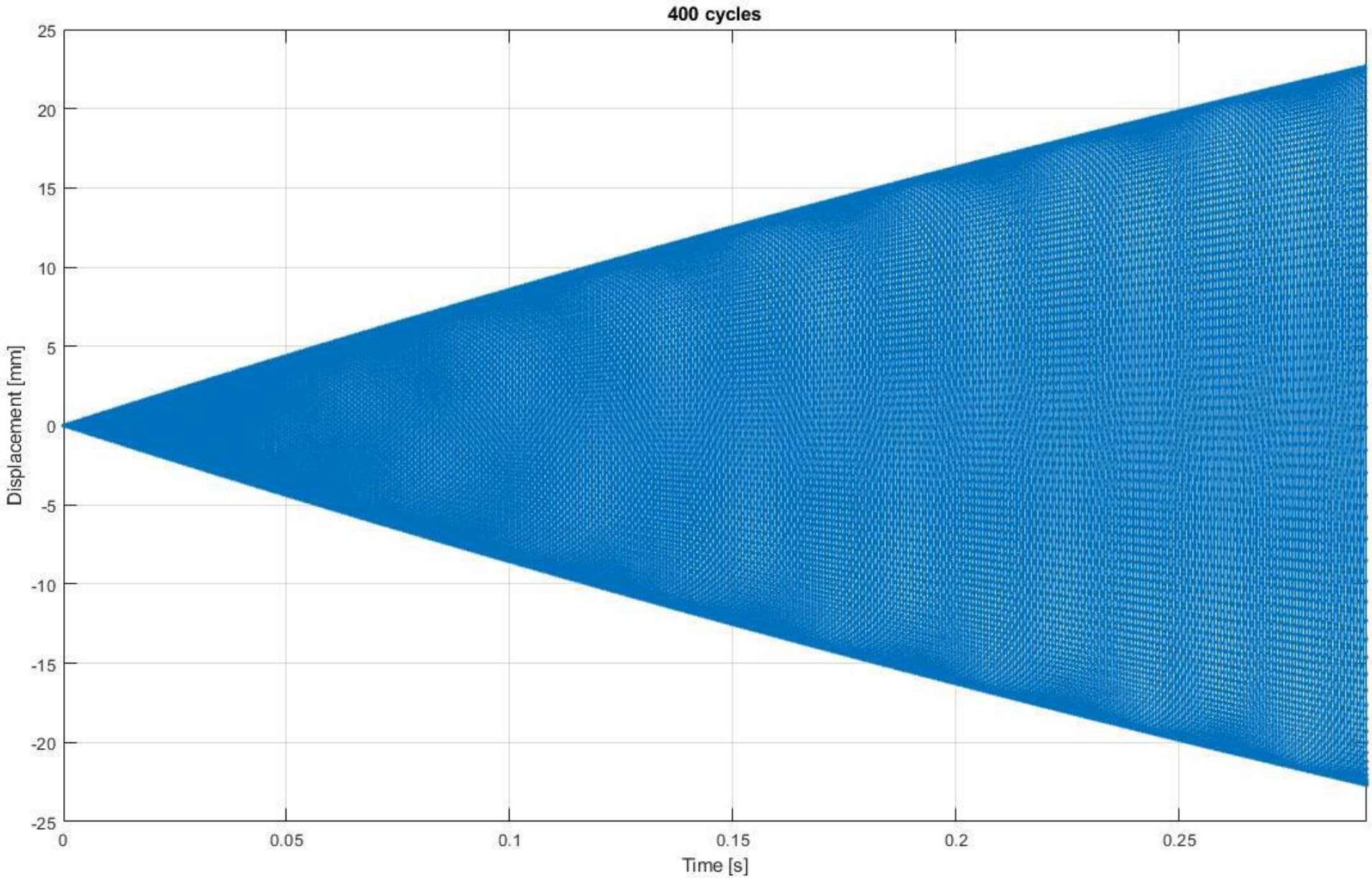
Plot results at the same manner as during first transient analysis (slides 17-18).






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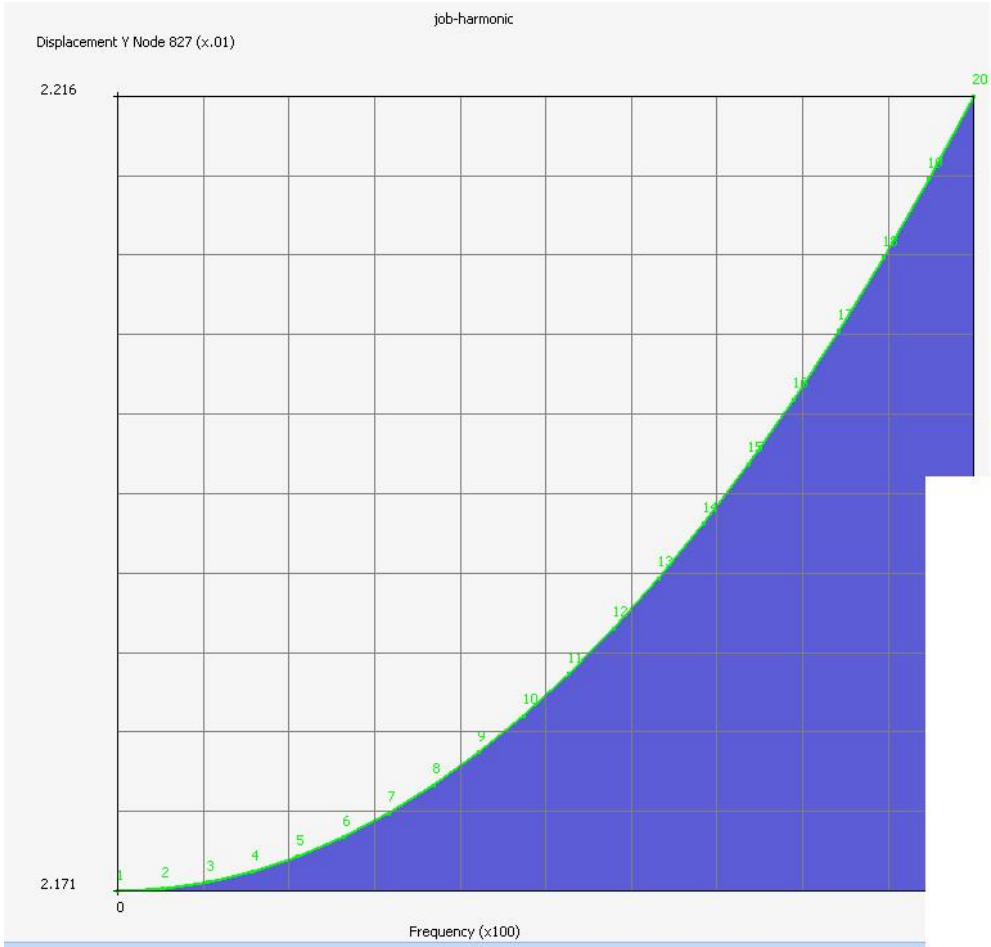
# 2nd problem: Transient analysis



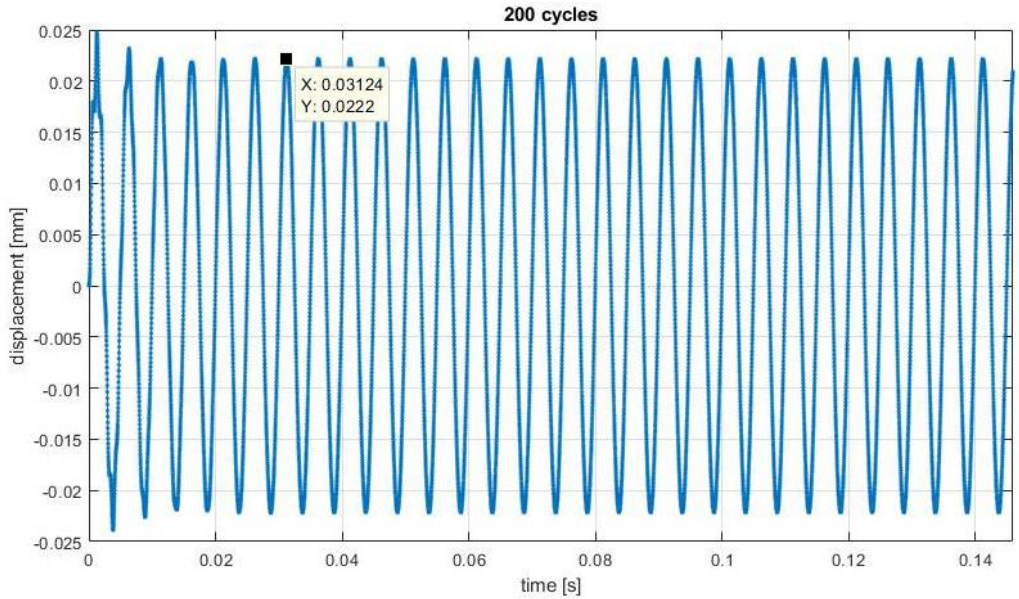



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# Harmonic vs Transient Analysis




The harmonic analysis can be used to estimate frequency of an harmonic load which gurantee demanded deflection (or estimate deflection under given load), while from transient analysis one can get when system becomes stable (at time domain).



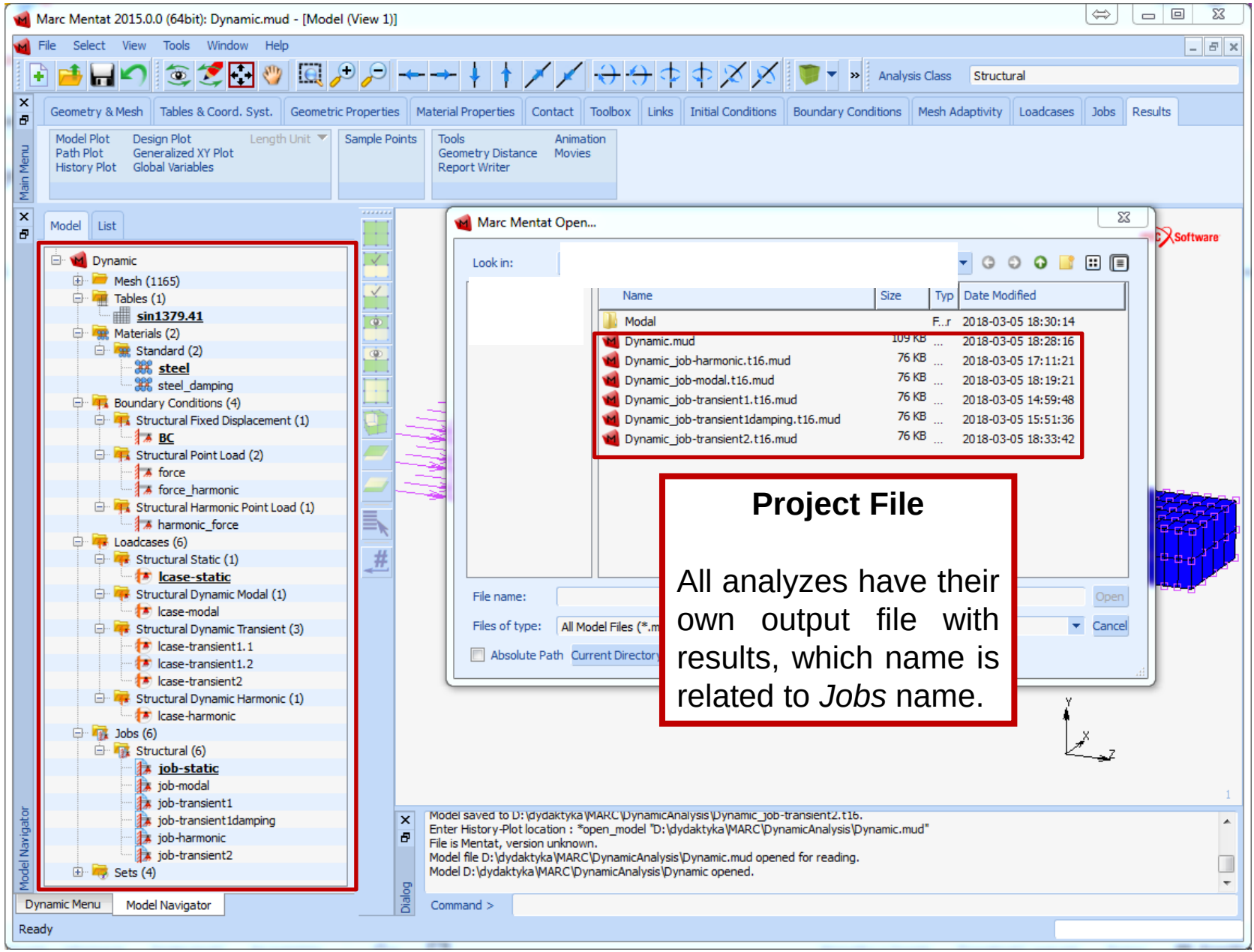



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# Project arrangement

**Project Tree**

All items of analyzes are divided according to analysis types.




**Project File**

All analyzes have their own output file with results, which name is related to *Jobs* name.